

Asymptotically Optimal Mixed-State Cloning

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Abstract

We study asymptotic cloning under root Uhlmann fidelity with the entire product output as target. In fixed dimension d , let n inputs and m_n outputs satisfy $m_n/n \rightarrow \gamma > 1$. For a known simple full-rank spectrum $p_1 > \cdots > p_d > 0$, we prove that the optimal fidelity on its unitary orbit is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}}, \quad q_{ij} = \frac{p_j}{p_i}.$$

When the spectrum is unknown, one spectrum-independent cloner attains

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} F_{\text{orb}}(\gamma, p),$$

uniformly on compact subsets of the simple full-rank chamber. This channel is minimax optimal on compact spectral sets whose interior in the full spectral hyperplane is dense in the set. The two factors quantify eigenbasis and spectral uncertainty. Our channels combine classical processing of Schur–Weyl labels with a covariant Cartan-sector channel; local asymptotic normality and a Gaussian fidelity bound against arbitrary channels yield the converses. For flat projector states $\rho = P/r$, with P a rank- r orthogonal projector, we separately prove the optimum $\gamma^{-r(d-r)/2}$. Computing the exact asymptotic fidelity of purify–clone–trace (PCT) at simple full-rank spectra, and bounding it from above for projector states, we show that our cloners strictly outperform PCT at every fixed gain on both families (for projectors, when $1 < r < d$).

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1 Introduction

The no-cloning theorem rules out perfect copying of an arbitrary unknown quantum state [WZ82, Die82]. Approximate cloning asks how well a channel can turn n identical inputs into m copies, at *gain* m/n . We compare the entire output with the product target $\rho^{\otimes m}$, rather than testing its marginals as in broadcasting [BCFJS96]. Optimal pure-state cloners were established in the 1990s [Wer98, KW99], but mixed-state cloning has remained open [SIGA05]: existing results treat restricted families, different loss criteria, or small-error sample complexity, without determining the optimal global fidelity in our setting [BDMP05, BDMP06, DF07, FLS07, Kaz10, LTHC26, FGSS26, JSO26]. We determine the known-spectrum orbit optimum and an unknown-spectrum compact-set minimax optimum for simple¹ full-rank spectra at every fixed asymptotic gain $m_n/n \rightarrow \gamma > 1$.

The eigenvalues and eigenbasis carry different kinds of information. When the spectrum is known, only the eigenbasis must be cloned. When it is unknown, the measured eigenvalues retain $n^{-1/2}$ -scale fluctuations that must be converted to the output scale. We construct cloners for both settings. The spectrum-independent cloner attains the known-spectrum optimum times a classical factor, uniformly on compact spectral sets. This value is minimax optimal when the set has dense full-dimensional interior (Theorem 1.2).

We also solve cloning of flat projector states $\rho = P/r$, where P is an unknown rank- r projector. We compare our cloners with purify-clone-trace (PCT) [LTHC26, FGSS26, JSO26]. Its exact asymptotic fidelity at simple full-rank spectra and an upper bound for projector states show that our cloners strictly outperform PCT on these families (for $1 < r < d$ in the projector case). The comparison concerns the protocols themselves, rather than their published lower guarantees.

Two complementary decompositions drive the solution. Schur–Weyl duality separates the spectral information in the Young label from the eigenbasis information in the conditional irreducible state [KW01]. Our cloners process the label classically and apply a multiplicity-one Cartan-sector channel to the irreducible state [PRV67, FH91]. Local asymptotic normality (LAN) yields a classical Gaussian shift and displaced thermal oscillator modes. The converse combines LAN with a Gaussian fidelity bound against arbitrary, possibly non-Gaussian and correlated channels. We prove the bound by a flat-prior argument; a product of least-noise amplifiers attains the quantum optimum. Projector states instead admit exact Schur-sector compression and a direct finite-dimensional converse.

1.1 Results

Fix an integer $d \geq 2$ and positive integers m_n such that

$$\gamma_n := \frac{m_n}{n} \longrightarrow \gamma > 1. \quad (1)$$

In particular, $m_n > n$ for all sufficiently large n .² Let $\mathbf{A}_d := \{x \in \mathbb{R}^d : \sum_i x_i = 1\}$ and define

$$\Delta_d^\circ := \{p \in \mathbf{A}_d : p_1 > \cdots > p_d > 0\}, \quad \rho_p = \text{diag}(p_1, \dots, p_d), \quad \rho_{p,g} = g\rho_p g^*. \quad (2)$$

¹A full-rank spectrum is *simple* (non-degenerate) if its eigenvalues are distinct.

²At the finitely many exceptional indices with $m_n \leq n$, our cloners are defined as the exact partial trace channel (and the identity when $m_n = n$). This convention does not affect any limit, and all construction formulas below concern $m_n > n$.

For positive trace-class operators A, C , not necessarily normalized, we write

$$F(A, C) = \|\sqrt{A}\sqrt{C}\|_1. \quad (3)$$

For states, this is the root (unsquared) Uhlmann fidelity. For nonnegative functions or measures, the same notation denotes their Hellinger affinity; in particular, $F(P, Q) = \sum_x \sqrt{P(x)Q(x)}$ on a countable space. The target in every statement below is the entire product state $\rho_{p,g}^{\otimes m_n}$. For $p \in \Delta_d^\circ$, define

$$q_{ij} = \frac{p_j}{p_i}, \quad f_\gamma(q) = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q}, \quad (4)$$

and

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_\gamma(q_{ij}), \quad F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2}, \quad F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p). \quad (5)$$

The factor F_{orb} measures the cost of amplifying the unknown eigenbasis. The *classical fidelity factor* F_{cl} accounts for the additional cost of reproducing the unknown spectral fluctuations.

Known spectrum. For fixed p , the target Young-label distribution is known and can be sampled exactly at size m_n ; only the conditional representation sectors encoding the eigenbasis need to be cloned.

Theorem 1.1 (Known-spectrum optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \text{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = F_{\text{orb}}(\gamma, p), \quad (6)$$

where the supremum is over all quantum channels from n to m_n qudits. The explicit sequence of channels $\mathcal{C}_n^p$ defined in [Section 2.2](#) asymptotically attains the displayed value.

Fixed-spectrum LAN transfers the Gaussian optimum for each displaced-thermal pair $i < j$ to the orbit; see [Section 3](#).

Unknown spectrum. Without p , the target Young-label law must instead be generated from the measured input label. Dilating its classical fluctuation to the output scale produces the factor $F_{\text{cl}}(\gamma, d)$.

Theorem 1.2 (Unknown-spectrum cloner fidelity and compact-set optimum). *There exists a sequence of spectrum-independent, $\text{U}(d)$ -covariant channels $\mathcal{C}_n^{\text{univ}}$, constructed in [Section 2.3](#), with the following properties.*

(a) *For every compact $K \subseteq \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\substack{p \in K \\ g \in \text{U}(d)}} \left| F(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) - F_{\text{univ}}(\gamma, p) \right| = 0. \quad (7)$$

(b) *Let $K \subseteq \Delta_d^\circ$ be nonempty and satisfy $K = \overline{\text{int}_{\mathbf{A}_d} K}^{\mathbf{A}_d}$. Then*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \text{U}(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \inf_{p \in K} F_{\text{univ}}(\gamma, p), \quad (8)$$

where the supremum is over all quantum channels from n to m_n qudits.

Part (a) evaluates one prescribed cloner, even on singleton or lower-dimensional sets. In part (b), the competing channel may depend on K , but not on the actual p or g . The interior is taken in the full $(d-1)$ -dimensional hyperplane $\mathbf{A}_d$: its spectral directions give the converse factor $F_{\text{cl}}(\gamma, d)$, and density extends the bound to the infimum over K ; see [Section 3.4](#). The condition matters. For $K = \{p\}$, the known-spectrum optimum is $F_{\text{orb}}(\gamma, p) > F_{\text{univ}}(\gamma, p)$, although part (a) still describes the spectrum-independent cloner.

A natural example bounds the smallest eigenvalue and every adjacent gap away from zero. For $0 < \kappa < 2/[d(d+1)]$, take

$$\Delta_{d,\kappa} = \{p \in \mathbf{A}_d : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa \ (i < d)\}.$$

This is a nonempty compact subset of Δ_d° and satisfies $\Delta_{d,\kappa} = \overline{\text{int}_{\mathbf{A}_d} \Delta_{d,\kappa}}^{\mathbf{A}_d}$, so (8) applies.³

[Figure 1](#) summarizes the common construction. The Schur transform separates a Young label, carrying spectral information, from a conditional irreducible state, carrying eigenbasis information. Each cloner chooses an output label and applies the same covariant sector channel; only the label transition rule changes. We develop these rules and their fidelities in [Section 2](#).

Theorem 1.3 (Cloning projector states). *For fixed $1 \leq r < d$, let $\text{Gr}(r, d)$ denote the set of rank- r orthogonal projectors on $\mathbb{C}^d$. Then*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{P \in \text{Gr}(r, d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) = \gamma^{-r(d-r)/2}. \quad (9)$$

The limit is attained by our cloner $\mathcal{C}_n^{\text{proj}}$ constructed in [Section 4.2](#).

For $r = 1$, the value $\gamma^{-(d-1)/2}$ recovers the optimal asymptotic global fidelity for pure-state cloning [[Wer98](#)]; for $r = d$, the family contains only I_d/d , and the value is one. The projector cloner uses the same Schur–Cartan architecture, with a different Young-label transition rule ([Fig. 1](#)). Its classical factor is 1 and its asymptotic sector factor is $\gamma^{-r(d-r)/2}$. Both bounds are proved separately in [Section 4](#).

1.2 Scope and limitations

Apart from flat projector states, our theorems require a simple full-rank spectrum. We do not treat nonflat rank-deficient spectra, positive degeneracies, or neighborhoods in which the rank changes. Our asymptotic estimates are established at fixed gain $\gamma > 1$; we do not control the joint regime in which the sample size grows and $\gamma \downarrow 1$. We also do not determine the exact finite-copy optimum. Expected extensions to degenerate and rank-deficient spectra, and the noncommuting limits at degeneracies, are discussed in [Section 5.4](#).

1.3 Earlier and related work

Pure-state cloning. Universal qubit cloners [[BH96](#), [GM97](#), [BDE+98](#)] were followed by Werner’s solution of global universal cloning in arbitrary dimension [[Wer98](#)] and Keyl–Werner’s optimality result for individual-clone tests [[KW99](#)]. State-dependent global cloning was developed in [[BDE+98](#), [CB99](#)]. The many-output limit for individual clones is tied to estimation [[MP95](#), [BEM98](#), [BA06](#), [CD06](#)]. For joint fidelity, optimal measure-and-prepare outputs need not be independent copies of

³Being convex with nonempty relative interior, $\Delta_{d,\kappa}$ is the closure of that interior. For a relative interior point, choose $\kappa' \in (\kappa, 2/[d(d+1)])$ and set $p_i = 1/d + ((d+1)/2 - i)\kappa'$. At the endpoint $\kappa = 2/[d(d+1)]$, the constraints force a singleton, so the compact-set theorem no longer applies.

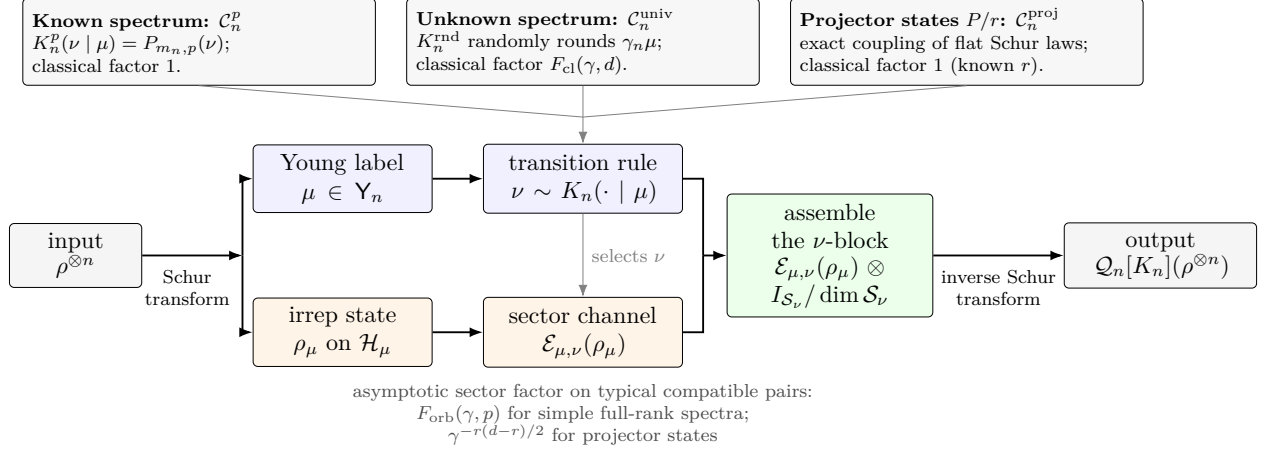


Figure 1: Schur–Cartan cloner architecture. A Young-label transition rule selects ν , while the covariant channel $\mathcal{E}_{\mu, \nu}$ processes the conditional irrep state. Known-spectrum, unknown-spectrum, and projector cloners differ only in their label rules, shown above; the listed classical and sector factors give their asymptotic fidelities in [Theorems 1.1 to 1.3](#).

an estimate [\[CY14\]](#). This distinction also matters for coherent-state and phase-covariant cloning [\[CKNWW05, KC22\]](#): Cerf et al. prove Gaussian optimality for joint fidelity, although non-Gaussian operations can improve single-clone fidelity. Reviews include [\[SIGA05, Chi11, FWJ+14\]](#).

Mixed-state cloning, broadcasting, and PCT. Exact broadcasting, and hence cloning, of a noncommuting family is impossible [\[BCFJS96, KH08\]](#). Approximate results include global-fidelity, relative-error, and state-dependent bounds for finite ensembles [\[Ras03a, Ras03b, Ras03c\]](#); cloners for symmetric-subspace inputs and identical mixed qubits under one-site shrinking-factor criteria [\[Fan03, FLS07\]](#); a restricted mixed-qubit construction under averaged Hilbert–Schmidt loss [\[Kaz10\]](#); and entropic limitations on simultaneous approximate cloning and broadcasting [\[LW17\]](#). Cirac–Ekert–Macchiavello solved mixed-qubit purification and a cloning/estimation problem under local fidelity [\[CEM99\]](#). Multi-input broadcasting and superbroadcasting also optimize marginals and permit correlations [\[DMP05, BDMP05, BDMP06, DF07\]](#); these criteria do not determine fidelity with the joint product target.

Random purification [\[TWZ25\]](#) followed by pure-state cloning gives a global-fidelity guarantee for bounded-rank mixed states [\[LTHC26\]](#). Concurrent work shows that PCT attains the optimal sample complexity up to constants for squared fidelity $1 - \epsilon$, uniformly over states of rank at most r [\[FGSS26, JSO26\]](#). These worst-case high-fidelity results use a different order of limits from our exact fixed-gain values; see [Section 5.3](#).

Gaussian cloning and LAN. Guță–Matsumoto solved cloning of displaced thermal oscillator states with known temperature under joint norm-distance loss, reducing it to amplification and proving Gaussian optimality by stochastic ordering [\[GM06\]](#). Bowles–Guță–Adesso used LAN for asymptotically optimal mixed-qubit purification and dilution under global trace-norm loss [\[BGA11\]](#); related methods solve qudit channel reversal [\[BGA12\]](#). These norm-distance results do not determine our fidelity constants, separate spectral and orbital costs, or finite-dimensional cloners.

The relevant LAN framework was developed in [\[GJ07, GK06, KG09, YFG13\]](#). Our fidelity amplification bound is attained by the least-noise amplifier [\[Cav82, GM06\]](#). Low-rank LAN has a

different mode structure [LN24]; we treat the boundary directly only for flat projector states.

1.4 Organization

Section 2 proves achievability for the Schur–Cartan cloners, using the sector and label estimates of Appendices B and C. Section 3 proves full-rank optimality via LAN and the Gaussian converse of Appendix D. Section 4 treats projector states independently. Section 5 compares PCT (analyzed in Appendix A), discusses sample complexity, and states conjectures and open problems.

Glossary of notation

We use $U(d)$ for the unitary group and $\mathcal{T}_1(\mathcal{H})$ for trace-class operators. The dimension d is fixed, $\gamma_n = m_n/n \rightarrow \gamma > 1$ gives the gain and its limit, N is a generic sample size, and $r_d = \binom{d}{2}$. Fidelity is unsquared: $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$. Unless stated otherwise, limits are as $n \rightarrow \infty$ with d, γ fixed, and $o(1)$ is uniform only over the parameters specified in its statement.

Symbols	Meaning
$A_d, \Delta_d^\circ; K \subseteq \Delta_d^\circ$	The affine hyperplane $\sum_i p_i = 1$, its strictly ordered positive chamber, and a compact subset of that chamber. Relative interiors and closures are in A_d .
$\rho_p, \rho_{p,g}; P \in \text{Gr}(r, d)$	$\rho_p = \text{diag}(p)$, $\rho_{p,g} = g\rho_p g^*$; a rank- r orthogonal projector, giving the state P/r , $1 \leq r < d$.
$q_{ij}, f_\gamma(q_{ij})$	$q_{ij} = p_j/p_i$ and its one-mode fidelity factor (4).
$F_{\text{orb}}, F_{\text{cl}}, F_{\text{univ}}$	Orbit optimum, classical fidelity factor, and their product $F_{\text{univ}} = F_{\text{cl}}F_{\text{orb}}$, attained uniformly on compact spectral sets; see (5) and the minimax statement in Theorem 1.2.
$\Upsilon_N; \mathcal{H}_\lambda, \mathcal{S}_\lambda$	Partitions $\lambda \vdash N$ with at most d rows; the associated unitary and symmetric-group representation spaces.
$P_{N,p}, \rho_{p,\lambda}; \mathbb{T}_N(p)$	Young-label law of $\rho_p^{\otimes N}$, conditional irreducible state, and typical-label set.
$K_n(\nu \mid \mu); \mathcal{Q}_n[K_n]$	A transition rule on Young labels and its covariant channel lift. The spectral set K is distinct from the transition rule K_n .
$\mathcal{C}_{\mu,\nu}, \mathcal{E}_{\mu,\nu}$	Cartan-sector channel for compatible labels and its extension to all label pairs by a fallback channel.
$\mathcal{M}_n; \mathcal{C}_n^p, \mathcal{C}_n^{\text{univ}}, \mathcal{C}_n^{\text{proj}}$	A general cloner; our known-spectrum, unknown-spectrum, and projector-state cloners.
$\mathbf{H}; h, z$	Spectral tangent hyperplane $\sum_i h_i = 0$; local spectral and orbital coordinates $h \in \mathbf{H}$, $z \in \mathbb{C}^{r_d}$.
$Y_p(z), g_{n,z}^{(p)}; \Omega_L$	Skew-Hermitian generator at the fixed base spectrum p , $g_{n,z}^{(p)} = e^{Y_p(z)/\sqrt{n}}$; bounded local parameter box.
$\mathcal{F}, \mathcal{F}_{\text{orb}}; \hat{N}$	One-mode Fock space, $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$, and number operator; distinct from fidelity F and sample size N .
$\tau_p, \tau_p^{(\gamma)}$	Thermal product in the input and ideal target, and noisy amplifier output at zero displacement.
$\Phi_z^{(p)}, \Psi_z^{(p,\gamma)}; \Theta^{\text{in}}, \Theta^{\text{tar}}$	Displaced input and ideal target built from τ_p ; their counterparts including the classical Gaussian component.

Symbols confined to one proof are defined there.

2 Cloner construction and fidelity analysis

The known- and unknown-spectrum protocols share the architecture in Fig. 1: a transition rule on Young labels lifts to a covariant quantum channel. We construct the two rules and combine their label estimates with a common sector analysis to evaluate the fidelities in Section 2.5.

Let $\mathsf{Y}_N = \{\lambda \vdash N : \ell(\lambda) \leq d\}$. Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\lambda \in \mathsf{Y}_N} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda, \quad (10)$$

where $\mathcal{H}_\lambda$ is the irreducible $U(d)$ -module of highest weight λ and $\mathcal{S}_\lambda$ is the corresponding Specht module. For $\rho_p = \text{diag}(p_1, \dots, p_d)$,

$$\rho_p^{\otimes N} = \bigoplus_{\lambda \in \mathsf{Y}_N} P_{N,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (11)$$

Here π_λ is the irreducible polynomial representation of highest weight λ , acting on $\mathcal{H}_\lambda$, and $s_\lambda(p) = \text{Tr } \pi_\lambda(\rho_p)$ is the corresponding Schur polynomial. For $\rho_{p,g} = g\rho_p g^*$, replace $\rho_{p,\lambda}$ by $\rho_{p,g,\lambda} = U_\lambda(g)\rho_{p,\lambda}U_\lambda(g)^*$, where $U_\lambda(g) = \pi_\lambda(g)$. The Young law $P_{N,p}$ depends on p , but not on g .

Define the typical Young-label set

$$\mathsf{T}_N(p) = \left\{ \lambda \in \mathsf{Y}_N : \left\| \frac{\lambda}{N} - p \right\|_\infty \leq N^{-1/3} \right\}. \quad (12)$$

Lemma 2.1 (Uniform Young concentration). *For all sufficiently large N , depending only on d ,*

$$\sup_{\substack{p_1 \geq \dots \geq p_d > 0 \\ \sum_{i=1}^d p_i = 1}} P_{N,p}(\mathsf{T}_N(p)^c) \leq e^{-N^{1/3}/4}. \quad (13)$$

Equal positive eigenvalues are allowed.

Proof. Write $q = \lambda/N$. The Weyl dimension formula gives $\dim \mathcal{H}_\lambda \leq (N+1)^{d(d-1)/2}$. Every weight ξ of $\mathcal{H}_\lambda$ has $\lambda - \xi$ equal to a nonnegative integral combination of positive roots. Since $p_j/p_i \leq 1$ for $i < j$, this gives $p^\xi \leq p^\lambda$. Hence $s_\lambda(p) \leq \dim \mathcal{H}_\lambda p^\lambda$. The standard bounds $\dim \mathcal{S}_\lambda \leq \binom{N}{\lambda_1, \dots, \lambda_d}$ and $\binom{N}{\lambda_1, \dots, \lambda_d} p^\lambda \leq \exp[-ND(q\|p)]$ give

$$P_{N,p}(\lambda) \leq (N+1)^{d(d-1)/2} \exp[-ND(q\|p)], \quad (14)$$

which is the estimate underlying the Keyl–Werner spectrum large-deviation theorem [KW01].

There are at most $(N+1)^{d-1}$ partitions of N with at most d rows. If $\lambda \notin \mathsf{T}_N(p)$, then $\|q - p\|_1 > N^{-1/3}$, and Pinsker’s inequality gives $D(q\|p) \geq \frac{1}{2}N^{-2/3}$. Therefore

$$P_{N,p}(\mathsf{T}_N(p)^c) \leq (N+1)^{d(d-1)/2+d-1} e^{-N^{1/3}/2}.$$

For all sufficiently large N , the polynomial factor is at most $e^{N^{1/3}/4}$, which proves (13). $\square$

2.1 Cartan-sector channel and lifting of a transition rule

Let λ, θ be dominant weights, with unit highest-weight vectors v_λ, v_θ . The Cartan component $\mathcal{H}_{\lambda+\theta}$ occurs with multiplicity one in $\mathcal{H}_\lambda \otimes \mathcal{H}_\theta$. Let

$$V_{\lambda,\theta} : \mathcal{H}_{\lambda+\theta} \longrightarrow \mathcal{H}_\lambda \otimes \mathcal{H}_\theta, \quad V_{\lambda,\theta} v_{\lambda+\theta} = v_\lambda \otimes v_\theta, \quad (15)$$

be the normalized intertwining isometry. For $\mu \in \mathbf{Y}_n$ and $\nu \in \mathbf{Y}_m$, write $\nu - \mu \vdash m - n$ when the coordinatewise difference $\nu - \mu$ is a partition of $m - n$, and call such a pair (μ, ν) *compatible*. If this holds, put $\omega = \nu - \mu$ and define

$$\mathcal{C}_{\mu,\nu}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^* (X \otimes I_{\mathcal{H}_\omega}) V_{\mu,\omega}. \quad (16)$$

The partial trace $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*)$ commutes with the irreducible $\text{U}(d)$ -action on $\mathcal{H}_\mu$; hence Schur's lemma and a trace comparison give $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*) = \frac{\dim \mathcal{H}_\nu}{\dim \mathcal{H}_\mu} I_{\mathcal{H}_\mu}$. The map in (16) is completely positive; the identity above makes it trace preserving, and equivariance of $V_{\mu,\omega}$ gives $\text{U}(d)$ -covariance.

When $\mu = (n, 0, \dots, 0)$ and $\nu = (m, 0, \dots, 0)$, the three irreps in (16) are symmetric, and the formula is Werner's universal pure-state n -to- m cloner [Wer98]. Thus the Cartan-sector channel extends Werner's construction from symmetric tensors to arbitrary compatible Schur sectors.

For every pair $(\mu, \nu) \in \mathbf{Y}_n \times \mathbf{Y}_m$, define a covariant channel $\mathcal{E}_{\mu,\nu} : \mathcal{T}_1(\mathcal{H}_\mu) \rightarrow \mathcal{T}_1(\mathcal{H}_\nu)$ by

$$\mathcal{E}_{\mu,\nu}(X) = \begin{cases} \mathcal{C}_{\mu,\nu}(X), & \nu - \mu \vdash m - n, \\ \text{Tr}(X) I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu, & \text{otherwise.} \end{cases} \quad (17)$$

Let $K_n : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$ be any transition rule: for each input label μ , the output label ν is selected with probability $K_n(\nu \mid \mu) \geq 0$, where $\sum_\nu K_n(\nu \mid \mu) = 1$. Its Schur–Cartan lifting $\mathcal{Q}_n[K_n]$ acts as follows: apply the Schur transform, measure μ , discard the input Specht factor, sample ν from $K_n(\cdot \mid \mu)$, apply $\mathcal{E}_{\mu,\nu}$, adjoin $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$, and apply the inverse Schur transform. Each step is completely positive and trace preserving, so their composition is a channel. Because the label measurement and sampling are invariant and every map $\mathcal{E}_{\mu,\nu}$ is covariant, $\mathcal{Q}_n[K_n]$ is $\text{U}(d)$ -covariant.

Proposition 2.2 (Fidelity of a lifted transition rule). *Fix $p \in \Delta_d^\circ$ and $g \in \text{U}(d)$. Let*

$$\Pi_{n,p}(\mu, \nu) = P_{n,p}(\mu) K_n(\nu \mid \mu), \quad Q_{m_n,p}(\nu) = \sum_\mu \Pi_{n,p}(\mu, \nu), \quad (18)$$

and, for $\nu \in \mathbf{Y}_{m_n}$, let

$$A_{n,p,\nu} = \sum_{\mu \in \mathbf{Y}_n} \Pi_{n,p}(\mu, \nu) \mathcal{E}_{\mu,\nu}(\rho_{p,\mu}) \quad (19)$$

be the unnormalized irreducible output block, so that $\text{Tr } A_{n,p,\nu} = Q_{m_n,p}(\nu)$. The lifted channel has fidelity

$$F(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \sum_{\nu \in \mathbf{Y}_{m_n}} \sqrt{P_{m_n,p}(\nu)} F(A_{n,p,\nu}, \rho_{p,\nu}). \quad (20)$$

Proof. By covariance of $\mathcal{Q}_n[K_n]$ and unitary invariance of fidelity, we may take $g = I$. Under the output Schur decomposition, the actual and target states are

$$\bigoplus_{\nu \in \mathbf{Y}_{m_n}} A_{n,p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad \bigoplus_{\nu \in \mathbf{Y}_{m_n}} P_{m_n,p}(\nu) \rho_{p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}.$$

Fidelity is additive over orthogonal direct sums, multiplicative over tensor products, and satisfies $F(A, cB) = \sqrt{c} F(A, B)$ for $c \geq 0$. Since $F(I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu, I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu) = 1$, this proves (20). $\square$

2.2 Known spectrum: exact target-label sampling

Fix $p \in \Delta_d^\circ$ and define

$$K_n^p(\nu \mid \mu) = P_{m_n, p}(\nu). \quad (21)$$

Set

$$\mathcal{C}_n^p := \mathcal{Q}_n[K_n^p]. \quad (22)$$

The replacement branch in (17) leaves the sampled label unchanged, so the output Young law on $\rho_{p, g}^{\otimes n}$ is $P_{m_n, p}$ for every n and g .

2.3 Unknown spectrum: randomized rounding

For $N \in \mathbb{N}$, let

$$\mathbf{L}_N = \left\{ x \in \mathbb{Z}^d : \sum_i x_i = N \right\}.$$

Set $\nu_n^{\text{fb}} = (m_n, 0, \dots, 0)$. Given $\mu \in \mathbf{Y}_n$, draw independent variables $U_1, \dots, U_{d-1}$, each uniform on $[-1/2, 1/2)$, and set

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + U_i) + \frac{1}{2} \right\rfloor \quad (1 \leq i < d), \quad \tilde{\nu}_d = m_n - \sum_{i < d} \tilde{\nu}_i. \quad (23)$$

Thus $\tilde{\nu} \in \mathbf{L}_{m_n}$; let $D_n(\tilde{\nu} \mid \mu)$ denote its distribution. This is a spectrum-independent lattice realization of the dilation $\mu \mapsto \gamma_n \mu$. The uniform shifts avoid lattice bias in the Gaussian limit. Retain the rounded label when compatible, and otherwise use the fallback:

$$\nu = \begin{cases} \tilde{\nu}, & \tilde{\nu} - \mu \vdash m_n - n, \\ \nu_n^{\text{fb}}, & \text{otherwise.} \end{cases} \quad (24)$$

Compatibility implies $\tilde{\nu} \in \mathbf{Y}_{m_n}$, so the conditional law of ν defines a transition rule $K_n^{\text{rnd}} : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$. Define

$$\mathcal{C}_n^{\text{univ}} := \mathcal{Q}_n[K_n^{\text{rnd}}]. \quad (25)$$

Since the transition rule is independent of p , this is a spectrum-independent, $U(d)$ -covariant cloner.

2.4 Sector and label asymptotics

The following proposition gives the common sector limit, including the uniformity under mixing needed when several input labels lead to one output label.

Proposition 2.3 (Uniform Cartan-sector fidelity). *Let $K \in \Delta_d^\circ$ and let $\delta_n \geq 0$ satisfy $\delta_n \rightarrow 0$. For all sufficiently large n , every $p \in K$, $\mu \in \mathbf{Y}_n$, and $\nu \in \mathbf{L}_{m_n}$ satisfying*

$$\left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \quad \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n$$

also satisfies $\nu - \mu \vdash m_n - n$; in particular, $\nu \in \mathbf{Y}_{m_n}$. Moreover,

$$F(\mathcal{C}_{\mu, \nu}(\rho_{p, \mu}), \rho_{p, \nu}) = F_{\text{orb}}(\gamma, p) + o(1) \quad (26)$$

uniformly over all such p, μ, ν . More generally, for fixed p, ν as above and any probability distribution $(a_\mu)_\mu$ supported on input labels satisfying the same closeness condition,

$$F\left(\sum_\mu a_\mu C_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}\right) = F_{\text{orb}}(\gamma, p) + o(1), \quad (27)$$

uniformly over these distributions as well as p, ν .

The mechanism is already visible on states obtained by a fixed number of lowering operations. Writing $\omega = \nu - \mu$, the Cartan intertwiner satisfies

$$V_{\mu,\omega} F_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)}) V_{\mu,\omega},$$

where $F_\alpha^{(\lambda)}$ is the lowering operator for $\alpha = e_i - e_j$ on $\mathcal{H}_\lambda$. After normalization, each excitation splits binomially between the two factors, with fraction

$$s_{\alpha,n} = \frac{\mu_i - \mu_j}{\nu_i - \nu_j} = \frac{1}{\gamma} + o(1)$$

entering the μ -factor. Thus the conditional thermal parameter $q_{ij} = p_j/p_i$ is transformed into

$$1 - s_{\alpha,n} + s_{\alpha,n} q_{ij} \longrightarrow 1 - \frac{1 - q_{ij}}{\gamma} =: q_{ij}^{(\gamma)}. \quad (28)$$

Write $\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|$ for the thermal state in the number basis on the one-mode Fock space of [Section 3.2](#), and put

$$\tau_p = \bigotimes_{i < j} \tau_{q_{ij}}, \quad \tau_p^{(\gamma)} = \bigotimes_{i < j} \tau_{q_{ij}^{(\gamma)}}. \quad (29)$$

The low-weight matrices of the sector output and ideal sector therefore approach those of $\tau_p^{(\gamma)}$ and τ_p , whose fidelity is $F_{\text{orb}}(\gamma, p)$. The output coordinates depend on ν , not on μ , so the same state limit holds uniformly under mixing. The exact PBW calculation and uniform thermal-tail argument are given in [Appendix B](#).

The second estimate concerns only K_n^{rnd} .

Proposition 2.4 (Randomized Young-label dilation). *There is a constant $c_{\text{rnd}} < \infty$, depending only on d and the fixed sequence (γ_n) , such that, for every compact $K \Subset \Delta_d^\circ$ and all sufficiently large n , every $p \in K$ and $\mu \in \mathbb{T}_n(p)$ satisfy*

$$\text{supp } K_n^{\text{rnd}}(\cdot \mid \mu) \subseteq \{\nu \in \mathbb{Y}_{m_n} : \nu - \mu \vdash m_n - n, \quad \|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}\}. \quad (30)$$

For every such K , the induced output-label law $Q_{m_n,p}$ from [\(18\)](#) with $K_n = K_n^{\text{rnd}}$,

$$F(Q_{m_n,p}, P_{m_n,p}) = F_{\text{cl}}(\gamma, d) + o(1) \quad \text{uniformly for } p \in K. \quad (31)$$

The uniform shifts in [\(23\)](#) make the transition rule a discrete realization of continuous dilation. After centering at Np and scaling by $\sqrt{N}$, embed a label law as a piecewise-constant density on the fundamental cells of the label lattice. Before fallback replacement, the intersection-volume identity [\(146\)](#) identifies the embedded rounded law with the target-cell average of the $\sqrt{\gamma_n}$ -dilation of the embedded input law. Write

$$\Sigma_p := (\text{diag}(p) - pp^\top) \Big|_{\{x \in \mathbb{R}^d : \sum_i x_i = 0\}}. \quad (32)$$

The Young local limit is $\mathbf{N}_{0,\Sigma_p}$, so randomized rounding changes the limit to $\mathbf{N}_{0,\gamma\Sigma_p}$, whereas the ideal m_n -label law still converges to $\mathbf{N}_{0,\Sigma_p}$. Their fidelity (Hellinger affinity) is

$$F(\mathbf{N}_{0,\gamma\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{(d-1)/2} = F_{\text{cl}}(\gamma, d). \quad (33)$$

For typical input labels, every rounded output is compatible for all sufficiently large n ; fallback therefore changes only atypical inputs, whose total probability vanishes uniformly on compact spectral sets. The full proof is given in Appendix C.

2.5 Fidelity of the constructed cloners

We use the following continuity estimate for positive trace-class operators A, C, A', C' with trace at most one:

$$|F(A, C) - F(A', C')| \leq \|A - A'\|_1^{1/2} + \|C - C'\|_1^{1/2}. \quad (34)$$

Indeed, the Powers–Størmer inequality bounds $\|\sqrt{A} - \sqrt{A'}\|_2$ by $\|A - A'\|_1^{1/2}$, and similarly for C, C' . Adding and subtracting $\sqrt{A'}\sqrt{C}$, the reverse triangle inequality and Schatten Hölder give (34), since $\|\sqrt{C}\|_2, \|\sqrt{A'}\|_2 \leq 1$.

The sector-mixture estimate gives a common factorization for both cloners.

Lemma 2.5 (Factorization of the lifted fidelity). *Let $K_n : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$ be transition rules. Fix $K \subseteq \Delta_d^0$, and use the laws $\Pi_{n,p}, Q_{m_n,p}$ from (18). Suppose that, for some $\delta_n \rightarrow 0$, the set of close label pairs*

$$G_{n,p} = \left\{ (\mu, \nu) \in \mathbf{Y}_n \times \mathbf{Y}_{m_n} : \left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n \right\}$$

satisfies $\sup_{p \in K} \Pi_{n,p}(G_{n,p}^c) \rightarrow 0$. Then, uniformly for $p \in K$ and $g \in \mathbf{U}(d)$,

$$F\left(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}, \rho_{p,g}^{\otimes m_n})\right) = F_{\text{orb}}(\gamma, p)F(Q_{m_n,p}, P_{m_n,p}) + o(1). \quad (35)$$

Proof. By covariance, take $g = I$. Write $P = P_{m_n,p}$, $Q = Q_{m_n,p}$, and $G = G_{n,p}$. Omitting the maximally mixed Specht factors, the output and target are $O = \bigoplus_\nu A_{n,p,\nu}$ and $T = \bigoplus_\nu P(\nu)\rho_{p,\nu}$. For large n , all pairs in G are compatible by Proposition 2.3. Retain only these pairs:

$$O_G = \bigoplus_\nu A_\nu^G, \quad A_\nu^G = \sum_{\mu: (\mu, \nu) \in G} \Pi_{n,p}(\mu, \nu) \mathcal{C}_{\mu, \nu}(\rho_{p, \mu}), \quad Q_G(\nu) = \text{Tr } A_\nu^G.$$

Whenever $Q_G(\nu) > 0$, the mixture statement of Proposition 2.3 applies to $A_\nu^G/Q_G(\nu)$. The direct-sum fidelity identity (20) and $F(Q_G, P) \leq 1$ therefore give

$$|F(O_G, T) - F_{\text{orb}}(\gamma, p)F(Q_G, P)| \leq \eta_n, \quad \eta_n \rightarrow 0,$$

uniformly on K . Positivity and trace preservation give $\|O - O_G\|_1 = \|Q - Q_G\|_{\ell^1} = \Pi_{n,p}(G^c) =: \epsilon_{n,p}$. Applying (34) once to the states and once to the label laws yields

$$|F(O, T) - F_{\text{orb}}(\gamma, p)F(Q, P)| \leq \eta_n + 2\sqrt{\epsilon_{n,p}} = o(1),$$

which proves (35). □

We now complete the achievability proofs of Theorems 1.1 and 1.2.

Known spectrum. For K_n^p , the joint law is $P_{n,p} \otimes P_{m_n,p}$ and $Q_{m_n,p} = P_{m_n,p}$. With $\delta_n = n^{-1/3} \geq m_n^{-1/3}$, $G_{n,p}$ contains $\mathbb{T}_n(p) \times \mathbb{T}_{m_n}(p)$, so Young concentration makes the discarded mass vanish, while the label fidelity is exactly one. Thus (35), with $K = \{p\}$, gives

$$F\left(\mathcal{C}_n^p(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{orb}}(\gamma, p) + o(1),$$

proving achievability in Theorem 1.1.

Unknown spectrum. For K_n^{rnd} , take $\delta_n = n^{-1/3} + c_{\text{rnd}}/m_n$. By (30), every pair in the support of $\Pi_{n,p}$ with $\mu \in \mathbb{T}_n(p)$ lies in $G_{n,p}$, so the discarded mass is at most $P_{n,p}(\mathbb{T}_n(p)^c) = o(1)$ uniformly on K . Combining (35) and (31) yields

$$F\left(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) + o(1)$$

uniformly for $p \in K$ and $g \in \mathbb{U}(d)$, proving the achievability assertion of Theorem 1.2.

3 Optimality from LAN and Gaussian converses

We use two-way LAN to transport arbitrary cloners on local neighborhoods to channels between Gaussian models. After establishing the Gaussian bounds, we apply the transfer in Fig. 2 to complete the minimax proofs.

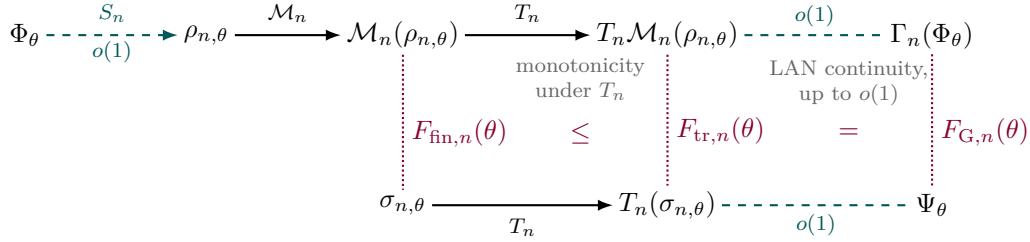


Figure 2: LAN transfer for $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$ on a fixed compact local parameter set C . The top and bottom rows track output and target; dotted segments indicate fidelity pairs, solid arrows exact channels, and dashed links uniform $o(1)$ trace-norm errors. Monotonicity and continuity give $F_{\text{fin},n}(\theta) \leq F_{\text{tr},n}(\theta) = F_{\text{G},n}(\theta) + o(1)$ uniformly on C , as proved in Lemma 3.5. The LAN maps need not be inverses, and Γ_n need not be Gaussian or converge.

3.1 Local coordinates

Fix a base spectrum $p \in \Delta_d^\circ$. Let

$$\mathbb{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad Y_p(z) = \sum_{j < k} \frac{z_{jk} E_{kj} - \bar{z}_{jk} E_{jk}}{\sqrt{p_j - p_k}}, \quad z \in \mathbb{C}^{r_d}, \quad r_d = \binom{d}{2},$$

where E_{jk} are the standard matrix units, and define

$$g_{n,z}^{(p)} = \exp\left(\frac{Y_p(z)}{\sqrt{n}}\right).$$

In $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}$, the coordinates h and z describe local changes of eigenvalues and eigenbasis, respectively. The denominators in Y_p use the fixed base spectrum p , even when the local spectrum is $p + h/\sqrt{n}$. For $L > 0$, set

$$\Omega_L = \{(h, z) \in \mathbf{H} \times \mathbb{C}^{r_d} : \|h\|_\infty \leq L, |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

For fixed p, L , $p + h/\sqrt{n} \in \Delta_d^\circ$ throughout Ω_L for large n . Under LAN, h becomes a classical Gaussian shift and z an oscillator displacement.

3.2 Gaussian amplification

Let $\mathcal{F}$ be the one-mode Fock space with number basis $\{|k\rangle : k \geq 0\}$, creation and annihilation operators a^*, a , and number operator $\hat{N} = a^*a$. We use the Weyl convention

$$D(z) = \exp(za^* - \bar{z}a). \quad (36)$$

For the thermal states τ_q introduced above, commutativity gives

$$F(\tau_q, \tau_{q'}) = \frac{\sqrt{(1-q)(1-q')}}{1 - \sqrt{qq'}}. \quad (37)$$

For $G \geq 1$, put $q^{(G)} = 1 - (1-q)/G$. The quantum-limited phase-insensitive amplifier $\mathcal{A}_G$ satisfies

$$\mathcal{A}_G(D(z)\tau_q D(z)^*) = D(\sqrt{G}z)\tau_{q^{(G)}} D(\sqrt{G}z)^*. \quad (38)$$

Equivalently, $\mathcal{A}_G(X) = \text{Tr}_b[S_r(X \otimes |0\rangle\langle 0|_b)S_r^*]$, where S_r is a two-mode squeezing unitary with $\cosh^2 r = G$ and the idler is in vacuum.

On $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$, use the thermal products $\tau_p, \tau_p^{(\gamma)}$ from (29) and put $D(z) = \bigotimes_{i < j} D(z_{ij})$. The three relevant states are

$$\begin{aligned} \text{input: } \quad & \Phi_z^{(p)} = D(z)\tau_p D(z)^*, \\ \text{ideal gain-}\sqrt{\gamma} \text{ target: } \quad & \Psi_z^{(p, \gamma)} = D(\sqrt{\gamma}z)\tau_p D(\sqrt{\gamma}z)^*, \\ \text{amplifier output: } \quad & \mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}) = D(\sqrt{\gamma}z)\tau_p^{(\gamma)} D(\sqrt{\gamma}z)^*. \end{aligned} \quad (39)$$

By (37),

$$F(\tau_{q^{(\gamma)}}, \tau_q) = f_\gamma(q). \quad (40)$$

Multiplicativity then gives, for every $z \in \mathbb{C}^{r_d}$,

$$F\left(\mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}), \Psi_z^{(p, \gamma)}\right) = F_{\text{orb}}(\gamma, p). \quad (41)$$

Equip $\mathbf{H}$ with its induced Euclidean measure and recall Σ_p from (32). Write $\mathbf{N}_{h, \Sigma_p}$ for the Gaussian law on $\mathbf{H}$ with mean h and covariance Σ_p . A classical-quantum state has the form

$$R(x) = r(x)\rho_x, \quad r(x) \geq 0, \quad \int_{\mathbf{H}} r(x) dx = 1,$$

where x is a classical variable, $r(x)$ is its density, and ρ_x is a quantum state on $\mathcal{F}_{\text{orb}}$. A classical-quantum channel Λ is a completely positive map, in the sense specified in Appendix D, that preserves $\int \text{Tr } R(x) dx$. If $S(x) = s(x)\sigma_x$, then

$$F(R, S) = \int_{\mathbf{H}} F(R(x), S(x)) dx = \int_{\mathbf{H}} \sqrt{r(x)s(x)} F(\rho_x, \sigma_x) dx.$$

This is the continuous direct-sum identity. With $\|R\|_1 := \int_{\mathbf{H}} \|R(x)\|_1 dx$, fidelity retains its monotonicity and joint-concavity properties.

Combining the classical spectral fluctuations with the quantum oscillator modes, define the input and ideal target states of the full Gaussian model by

$$\Theta_{h,z}^{\text{in}} = \mathbf{N}_{h,\Sigma_p} \otimes \Phi_z^{(p)}, \quad \Theta_{h,z}^{\text{tar}} = \mathbf{N}_{\sqrt{\gamma}h,\Sigma_p} \otimes \Psi_z^{(p,\gamma)}. \quad (42)$$

The target scales both displacement parameters (h, z) by $\sqrt{\gamma}$, while retaining the input covariance and thermal noise. For the orbital model, let Z_L be the box of oscillator displacements over which we average the fidelity:

$$Z_L = \{z \in \mathbb{C}^{r_d} : |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

Theorem 3.1 (Gaussian amplification optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F\left(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}\right) dz = F_{\text{orb}}(\gamma, p), \quad (43)$$

and

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F\left(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}\right) dh dz = F_{\text{univ}}(\gamma, p). \quad (44)$$

The first supremum is over all quantum channels on $\mathcal{F}_{\text{orb}}$; the second is over all classical-quantum channels. No Gaussianity, covariance, or product-structure assumption is imposed on the competing channels in either supremum.

The product amplifier $\mathcal{A}_\gamma^{\otimes r_d}$ attains $F_{\text{orb}}(\gamma, p)$ in the orbital model. Tensoring it with the classical pushforward $x \mapsto \sqrt{\gamma}x$ supplies the Gaussian affinity $F_{\text{cl}}(\gamma, d)$, attaining $F_{\text{univ}}(\gamma, p)$ in the full model.

For the converse, average a near-optimal channel over translates of the expanding flat-prior boxes and use a weighted-fidelity witness to reduce the problem to an oscillator moment. The weighted oscillator maps have a subsequential limit against compact observables that is completely positive and displacement covariant. Its output trace on normalized inputs is at most 1 in the orbital model and, by a Gaussian translation estimate, $F_{\text{cl}}(\gamma, d)$ in the full model. The least-noise inequality then gives the respective bounds $F_{\text{orb}}(\gamma, p)$ and $F_{\text{univ}}(\gamma, p)$; see [Proposition D.4](#) in [Appendix D](#).

3.3 Local asymptotic normality

Here LAN means that, uniformly for (h, z) in compact sets, the state families $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n}$ and $\mathbf{N}_{h,\Sigma_p} \otimes \Phi_z^{(p)}$ are asymptotically interconvertible by channels with vanishing trace-norm error. The Gaussian and oscillator factors represent spectral and eigenbasis fluctuations, respectively; cloning scales (h, z) by $\sqrt{\gamma}$.

We fix the oscillator normalization directly from the local tangent and the collective fluctuation operators. Put $\Delta_{ij} = p_i - p_j$. The chart in [Section 3.1](#) satisfies

$$\frac{[Y_p(z), \rho_p]_{ji}}{\sqrt{\Delta_{ij}}} = z_{ij}, \quad i < j. \quad (45)$$

For $a_{ij,n} := (n\Delta_{ij})^{-1/2} \sum_{s=1}^n E_{ij}^{(s)}$, where $E_{ij}^{(s)}$ acts on the s -th tensor factor, the expectations read

$$\left\langle [a_{ij,n}, a_{ij,n}^*] \right\rangle_{\rho_p^{\otimes n}} = 1, \quad \left\langle a_{ij,n}^* a_{ij,n} \right\rangle_{\rho_p^{\otimes n}} = \frac{p_j}{\Delta_{ij}} = \frac{q_{ij}}{1 - q_{ij}}.$$

Under the local state, (45) gives $\langle a_{ij,n} \rangle = z_{ij} + o(1)$, uniformly on compact parameter sets. Thus the limiting mode has mean z_{ij} and thermal parameter q_{ij} , agreeing with $D(z_{ij})\tau_{q_{ij}}D(z_{ij})^*$ in (36). The following two-way LAN statement follows from Kahn–Guță’s channel construction [KG09, Theorem 4.3, Lemma 6.4, and Section 7.3], in the oscillator coordinates fixed above and with the target-sample rescaling below.⁴

Theorem 3.2 (Finite-dimensional LAN and target-sample scaling). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{full}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \longrightarrow L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})), \quad S_N^{\text{full}} : L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})) \longrightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, for every compact $C \subset \mathbf{H} \times \mathbb{C}^d$,

$$\sup_{(h,z) \in C} \left\| T_n^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right) - \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right\|_1 \longrightarrow 0, \quad (46)$$

$$\sup_{(h,z) \in C} \left\| S_n^{\text{full}} \left(\mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right) - \rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right\|_1 \longrightarrow 0, \quad (47)$$

and

$$\sup_{(h,z) \in C} \left\| T_{m_n}^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes m_n} \right) - \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p, \gamma)} \right\|_1 \longrightarrow 0. \quad (48)$$

Coordinate identification and target scaling. For (46)–(47), identify the spectral coordinate $u \in \mathbb{R}^{d-1}$ of the cited theorem with $h = Au = (u_1, \dots, u_{d-1}, -\sum_{i < d} u_i) \in \mathbf{H}$. Its covariance $V(p)_{ij} = \delta_{ij}p_i - p_i p_j$, $i, j < d$ [KG09, Eq. (3.3)], satisfies $AV(p)A^\top = \text{diag}(p) - pp^\top$, giving $\mathbf{N}_{h, \Sigma_p}$; the normalization (45) identifies the oscillator factor with $\Phi_z^{(p)}$.

For (48), write $\gamma_n = m_n/n$ and use

$$\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}} = \rho_{p+\sqrt{\gamma_n}h/\sqrt{m_n}, g_{m_n, \sqrt{\gamma_n}z}^{(p)}}$$

with the same base p at both sample sizes. Apply (46) at sample size m_n on a common compact set containing $(\sqrt{\gamma_n}h, \sqrt{\gamma_n}z)$ for all sufficiently large n . Since $\gamma_n \rightarrow \gamma$, uniform trace-norm continuity of Gaussian translations and Weyl conjugation on that set gives the claimed target limit.

Remark 3.3 (Qubit normalization check). For $d = 2$, put $\Delta = p_1 - p_2 > 0$, $q = p_2/p_1$, and take $z = t \in \mathbb{R}$. The exact qubit fidelity identity gives

$$F\left(\rho_p^{\otimes n}, \rho_{p, g_{n,t}^{(p)}}^{\otimes n}\right) = \left(1 - \Delta^2 \sin^2 \frac{t}{\sqrt{n\Delta}}\right)^{n/2} \longrightarrow e^{-\Delta t^2/2}.$$

This equals $F(\tau_q, D(t)\tau_q D(t)^*) = \exp\left[-\frac{(1-q)t^2}{2(1+q)}\right]$, confirming the displacement normalization we use.

Corollary 3.4 (Fixed-spectrum LAN). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{orb}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \rightarrow \mathcal{T}_1(\mathcal{F}_{\text{orb}}), \quad S_N^{\text{orb}} : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \rightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, uniformly for z in compact sets,

$$\|T_n^{\text{orb}}(\rho_{p, g_{n,z}^{(p)}}^{\otimes n}) - \Phi_z^{(p)}\|_1 \longrightarrow 0, \quad (49)$$

⁴We use the displacement normalization of the constructive Lemma 6.4 and Section 7.3 of the cited work. The displayed Eq. (4.19) in its arXiv version has a different factor and adjoint placement. The rotation chart, fluctuation calculation, and qubit check below fix our convention.

$$\|S_n^{\text{orb}}(\Phi_z^{(p)}) - \rho_{p,g_{n,z}}^{\otimes n}\|_1 \longrightarrow 0, \quad (50)$$

and

$$\|T_{m_n}^{\text{orb}}(\rho_{p,g_{n,z}}^{\otimes m_n}) - \Psi_z^{(p,\gamma)}\|_1 \longrightarrow 0. \quad (51)$$

Proof. Restrict [Theorem 3.2](#) to $h = 0$ and set $T_N^{\text{orb}}(X) := \int_{\mathcal{H}} T_N^{\text{full}}(X)(x) dx$ and $S_N^{\text{orb}}(\sigma) := S_N^{\text{full}}(\mathbf{N}_{0,\Sigma_p} \otimes \sigma)$. Marginalization contracts trace norm, and tensoring with $\mathbf{N}_{0,\Sigma_p}$ preserves it, giving all three assertions. $\square$

Lemma 3.5 (LAN fidelity transfer). *Let $\{\rho_{n,\theta}\}_{\theta \in C}$ and $\{\sigma_{n,\theta}\}_{\theta \in C}$ be input and target models on a compact parameter set C . Let $\{\Phi_\theta\}$ and $\{\Psi_\theta\}$ be limiting models. Suppose channels S_n, T_n satisfy*

$$\begin{aligned} \delta_n &:= \sup_{\theta \in C} \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \rightarrow 0, \\ \eta_n &:= \sup_{\theta \in C} \|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \rightarrow 0. \end{aligned}$$

For every channel $\mathcal{M}_n$, put $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$. Then, for every $\theta \in C$,

$$F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) \leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}. \quad (52)$$

The error is uniform over both $\theta \in C$ and $\mathcal{M}_n$.

Proof. Trace-norm contractivity of $T_n \circ \mathcal{M}_n$ gives $\|\Gamma_n(\Phi_\theta) - T_n \mathcal{M}_n(\rho_{n,\theta})\|_1 \leq \delta_n$. Monotonicity under T_n , followed by [\(34\)](#) and the definition of η_n , therefore yields

$$\begin{aligned} F(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) &\leq F(T_n \mathcal{M}_n(\rho_{n,\theta}), T_n(\sigma_{n,\theta})) \\ &\leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}. \end{aligned}$$

$\square$

Fixed-label asymptotics versus LAN. The thermal states τ_p and $\tau_p^{(\gamma)}$ also appear in [Proposition 2.3](#), but there they are cutoff limits of the target and cloner-output sector states for fixed typical label pairs; see [\(143\)](#) and [\(142\)](#). LAN instead supplies channels between an entire local model and its Gaussian limit, and retains the label fluctuation as $\mathbf{N}_{h,\Sigma_p}$. The fixed-label calculation therefore gives achievability of $F_{\text{orb}}(\gamma, p)$, while LAN supplies the converse.

3.4 Completion of the minimax proofs

Known-spectrum converse and completion of Theorem 1.1. For the known-spectrum problem, set $u_n(p) := \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$. For every channel $\mathcal{M}_n$ and fixed L , set

$$\Gamma_n = T_{m_n}^{\text{orb}} \circ \mathcal{M}_n \circ S_n^{\text{orb}}.$$

The local orbit $\{g_{n,z}^{(p)} : z \in Z_L\}$ is contained in the full unitary orbit. Apply [Lemma 3.5](#), average over Z_L , and take the supremum over $\mathcal{M}_n$. Since the transfer error is independent of $\mathcal{M}_n$,

$$u_n(p) \leq \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}) dz + o(1),$$

where the supremum is over all channels on the Gaussian model and the error tends to zero for fixed L . Taking $\limsup_n$, then $L \rightarrow \infty$, [\(43\)](#) gives

$$\limsup_{n \rightarrow \infty} u_n(p) \leq F_{\text{orb}}(\gamma, p).$$

Together with $\liminf_n u_n(p) \geq F_{\text{orb}}(\gamma, p)$, proved in [Section 2](#), this proves [Theorem 1.1](#). $\square$

Fixing p would miss the cost of an unknown spectrum. Instead, we use $n^{-1/2}$ -neighborhoods of an interior spectrum and let the local window grow after the large- n limit.

Proof of part (b) of Theorem 1.2. Let $w_n(K)$ denote the quantity inside the limit on the left of (8). Part (a) gives $\liminf_n w_n(K) \geq \inf_{p \in K} F_{\text{univ}}(\gamma, p)$. For the reverse inequality, fix $p_0 \in \text{int}_{\mathbf{A}_d} K$ and L . For large n , all local spectra $p_0 + h/\sqrt{n}$, $(h, z) \in \Omega_L$, lie in K , so the worst-case fidelity of $\mathcal{M}_n$ on K is at most its average over this window. Repeating the known-spectrum argument with the full LAN maps at p_0 , $\Gamma_n = T_{m_n}^{\text{full}} \circ \mathcal{M}_n \circ S_n^{\text{full}}$, and Ω_L in place of $\mathbf{Z}_L$ gives

$$w_n(K) \leq \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F\left(\Lambda(\Theta_{h,z}^{\text{in}}, \Theta_{h,z}^{\text{tar}})\right) dh dz + o(1),$$

where both Gaussian models are based at p_0 and the supremum is over all classical-quantum channels. Taking $\limsup_n$ and then $L \rightarrow \infty$, (44) yields $\limsup_n w_n(K) \leq F_{\text{univ}}(\gamma, p_0)$. Since $\text{int}_{\mathbf{A}_d} K$ is dense in K and $F_{\text{univ}}(\gamma, \cdot)$ is continuous, taking the infimum over p_0 completes the proof. $\square$

4 Cloning projector states

A flat rank- r projector state has the form $\rho_P = P/r$, where P is a rank- r orthogonal projector. These projectors form the complex Grassmannian $\text{Gr}(r, d)$. Rank deficiency, and for $r > 1$ the repeated positive eigenvalue, place these states outside the simple full-rank theorems. We construct $\mathcal{C}_n^{\text{proj}}$ and prove Theorem 1.3 directly, without a general rank-deficient LAN theorem. The compression lemma below holds for any positive spectrum on its support, but does not by itself establish optimal cloning beyond the flat case.

4.1 Exact rank compression

In this section, a superscript (s) indicates the ambient dimension of a representation space or map. For a partition λ with at most r rows, the Weyl dimension formula gives

$$D_s^\lambda = \dim \mathcal{H}_\lambda^{(s)}, \quad R_\lambda = \frac{D_d^\lambda}{D_r^\lambda} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\lambda_i + j - i}{j - i}. \quad (53)$$

Let

$$J_\lambda : \mathcal{H}_\lambda^{(r)} \longrightarrow \mathcal{H}_\lambda^{(d)}$$

be the isometric $U(r)$ -intertwiner that maps the unit highest-weight vector to the unit highest-weight vector. Its range is the $U(r)$ -submodule generated by that vector, and we set $P_\lambda = J_\lambda J_\lambda^*$.

For a compatible pair of r -row partitions $\mu \vdash n$ and $\nu \vdash m$, write $\omega = \nu - \mu \vdash m - n$. In ambient dimension $s \in \{r, d\}$, the Cartan-sector channel is

$$\mathcal{C}_{\mu, \nu}^{(s)}(X) = \frac{D_s^\mu}{D_s^\nu} (V_{\mu, \omega}^{(s)})^* (X \otimes I_{\mathcal{H}_\omega^{(s)}}) V_{\mu, \omega}^{(s)}.$$

For a probability vector $x \in \mathbb{R}^s$, including one with zero entries, write $\pi_\lambda^{(s)}$ for the polynomial $U(s)$ -representation, $s_\lambda^{(s)}$ for its Schur polynomial, and

$$\rho_{x, \lambda}^{(s)} = \frac{\pi_\lambda^{(s)}(\text{diag } x)}{s_\lambda^{(s)}(x)}$$

whenever the denominator is nonzero.

Lemma 4.1 (Exact rank-compression identity). *Let $p = (p_1, \dots, p_r)$ satisfy $p_i > 0$ and $\sum_i p_i = 1$, and put $\bar{p} = (p_1, \dots, p_r, 0, \dots, 0) \in \mathbb{R}^d$. For every r -row partition λ ,*

$$\rho_{\bar{p}, \lambda}^{(d)} = J_\lambda \rho_{p, \lambda}^{(r)} J_\lambda^*. \quad (54)$$

For every compatible r -row pair μ, ν and every trace-class operator X on $\mathcal{H}_\mu^{(r)}$,

$$P_\nu \mathcal{C}_{\mu, \nu}^{(d)}(J_\mu X J_\mu^*) P_\nu = a_{\mu, \nu}^{(r, d)} J_\nu \mathcal{C}_{\mu, \nu}^{(r)}(X) J_\nu^*, \quad (55)$$

where

$$a_{\mu, \nu}^{(r, d)} = \frac{R_\mu}{R_\nu} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\mu_i + j - i}{\nu_i + j - i}. \quad (56)$$

Consequently, the fidelity for each pair of labels factorizes exactly:

$$F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{\bar{p}, \mu}^{(d)}), \rho_{\bar{p}, \nu}^{(d)}\right) = \sqrt{a_{\mu, \nu}^{(r, d)}} F\left(\mathcal{C}_{\mu, \nu}^{(r)}(\rho_{p, \mu}^{(r)}), \rho_{p, \nu}^{(r)}\right). \quad (57)$$

Proof. Let $\iota : \mathbb{C}^r \hookrightarrow \mathbb{C}^d$ be the coordinate inclusion. Schur–Weyl naturality identifies $\iota^{\otimes N}$ with $\bigoplus_{\ell(\lambda) \leq r} (J_\lambda \otimes I_{S_\lambda})$. Comparing blocks in $(\text{diag } \bar{p})^{\otimes N} = \iota^{\otimes N} (\text{diag } p)^{\otimes N} (\iota^*)^{\otimes N}$ gives

$$\pi_\lambda^{(d)}(\text{diag } \bar{p}) = J_\lambda \pi_\lambda^{(r)}(\text{diag } p) J_\lambda^*.$$

Taking traces gives $s_\lambda^{(d)}(\bar{p}) = s_\lambda^{(r)}(p)$; the longer-row Schur polynomials vanish. Dividing the operator identity by this common nonzero trace proves (54).

The two maps

$$V_{\mu, \omega}^{(d)} J_\nu, \quad (J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}$$

are $U(r)$ -intertwiners from $\mathcal{H}_\nu^{(r)}$ to $\mathcal{H}_\mu^{(d)} \otimes \mathcal{H}_\omega^{(d)}$. Both send the unit highest-weight vector to $v_\mu \otimes v_\omega$, so they agree on the submodule generated by that vector:

$$V_{\mu, \omega}^{(d)} J_\nu = (J_\mu \otimes J_\omega) V_{\mu, \omega}^{(r)}. \quad (58)$$

Inserting (58) into the two sector-channel formulas proves (55), with the dimension ratio in (56); its product form follows from (53).

If $A, B \geq 0$ and $PBP = B$, then $F(A, B) = \text{Tr} \sqrt{\sqrt{B} A \sqrt{B}} = F(PAP, B)$. Apply this with $P = P_\nu$. Then (54), (55), isometric invariance, and $F(aA, B) = \sqrt{a} F(A, B)$ give (57). $\square$

For the standard projector P_0 onto the first r coordinates, the flat-spectrum conditional state in the λ -sector is

$$\rho_{P_0, \lambda} = J_\lambda \frac{I_{\mathcal{H}_\lambda^{(r)}}}{D_r^\lambda} J_\lambda^*. \quad (59)$$

By $U(r)$ -covariance and irreducibility, the r -dimensional Cartan channel maps the maximally mixed input to the maximally mixed output. Thus (57) gives, for every compatible pair,

$$F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{P_0, \mu}), \rho_{P_0, \nu}\right) = \sqrt{\frac{R_\mu}{R_\nu}}. \quad (60)$$

For $\mu_i/n \rightarrow 1/r$ and $\nu_i/m_n \rightarrow 1/r$, this tends to the right-hand side of (9).

4.2 An asymptotically compatible coupling with exact marginals

Let

$$\mathbf{P}_N^{(r)}(\lambda) = \frac{\dim \mathcal{S}_\lambda D_r^\lambda}{r^N}, \quad \lambda \vdash N, \quad \ell(\lambda) \leq r, \quad (61)$$

be the Schur–Weyl law of the maximally mixed state on $\mathbb{C}^r$. The complete Schur decomposition is

$$(P/r)^{\otimes N} = \bigoplus_{\ell(\lambda) \leq r} \mathbf{P}_N^{(r)}(\lambda) \frac{\Pi_{\lambda, P}}{D_r^\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}. \quad (62)$$

Here $\Pi_{\lambda, P}$ is the rank- D_r^λ support projection defined intrinsically by the range of P . Padding partitions with zero rows to length r , the Schur–Weyl/random-word central limit theorem [Kup02, Joh01] states that

$$X_N := \frac{\lambda - (N/r)\mathbf{1}_r}{\sqrt{N}} \Rightarrow X, \quad (63)$$

where X is a fixed scaling of the ordered eigenvalue vector of a traceless $r \times r$ GUE matrix. In particular, $\Pr(X_1 > \dots > X_r) = 1$.

Both X_n and X_{m_n} converge to the same law. Their Prokhorov distance therefore tends to zero, so Strassen’s coupling theorem [Str65] supplies a coupling $\Pi_n(\mu, \nu)$ of the exact marginals $\mathbf{P}_n^{(r)}$ and $\mathbf{P}_{m_n}^{(r)}$ such that

$$\left\| \frac{\mu - (n/r)\mathbf{1}_r}{\sqrt{n}} - \frac{\nu - (m_n/r)\mathbf{1}_r}{\sqrt{m_n}} \right\|_\infty \rightarrow 0 \quad \text{in probability.} \quad (64)$$

Condition this finite coupling on the input label to obtain a transition rule $K_n(\nu \mid \mu)$, extending it arbitrarily to input diagrams with more than r rows. These have zero probability on projector inputs; the extension defines a channel on the entire input space. Set

$$\mathcal{C}_n^{\text{proj}} := \mathcal{Q}_n[K_n]. \quad (65)$$

Its dependence on the fixed rank r is suppressed in the notation; the channel is independent of the unknown projector P .

Put $\omega = \nu - \mu$. Its leading term is $(m_n - n)/r$ in every coordinate, while its fluctuations are of order $\sqrt{n}$. Hence $\omega_i > 0$ for every i with probability $1 - o(1)$. For adjacent rows,

$$\frac{\omega_i - \omega_{i+1}}{\sqrt{n}} = \sqrt{\gamma_n} \left[\frac{\nu_i - \nu_{i+1}}{\sqrt{m_n}} \right] - \left[\frac{\mu_i - \mu_{i+1}}{\sqrt{n}} \right] \Rightarrow (\sqrt{\gamma} - 1)(X_i - X_{i+1}) > 0.$$

Thus the compatibility event

$$C_n = \{(\mu, \nu) : \nu - \mu \vdash m_n - n\}$$

has probability $1 - o(1)$. The cloner $\mathcal{C}_n^{\text{proj}}$ uses the ambient Cartan-sector channel on C_n and, otherwise, the covariant replacement channel into the same sampled ν -block. The output-label marginal is exactly the target law. The central limit theorem also gives $\mu_i/n, \nu_i/m_n \rightarrow 1/r$ in probability, so (53) implies

$$\mathbf{1}_{C_n} \sqrt{\frac{R_\mu}{R_\nu}} \xrightarrow{\mathbb{P}} \gamma^{-r(d-r)/2}. \quad (66)$$

On C_n , compatibility gives $\nu_i \geq \mu_i$, hence $0 \leq \mathbf{1}_{C_n} \sqrt{R_\mu/R_\nu} \leq 1$. Thus (66) also holds in expectation. Since the output-label marginal is exact, the block identity (20) (valid also for r -row label laws),

monotonicity and concavity of F in its first argument, and (60) give

$$F\left(\mathcal{C}_n^{\text{proj}}((P/r)^{\otimes n}), (P/r)^{\otimes m_n}\right) \geq \mathbb{E}_{\Pi_n} \left[\mathbf{1}_{C_n} \sqrt{\frac{R_\mu}{R_\nu}} \right] \longrightarrow \gamma^{-r(d-r)/2}. \quad (67)$$

This proves achievability in [Theorem 1.3](#).

4.3 Direct finite-dimensional converse

Covariantize an arbitrary cloner over $U(d)$ and twirl its output over permutations. Neither decreases its worst-case fidelity: the first follows from joint concavity and unitary invariance, the second from monotonicity under the output twirl, which fixes every target. Write $\mathcal{M}_n$ for the resulting channel.

Replace each support projection in (62) by the identity to obtain the invariant positive operator

$$B_n = \bigoplus_{\substack{\mu \vdash n \\ \ell(\mu) \leq r}} \mathbf{P}_n^{(r)}(\mu) \frac{I_{\mathcal{H}_\mu^{(d)}}}{D_r^\mu} \otimes \frac{I_{\mathcal{S}_\mu}}{\dim \mathcal{S}_\mu}.$$

For every P ,

$$(P/r)^{\otimes n} \leq B_n, \quad \text{Tr } B_n = \sum_{\mu} \mathbf{P}_n^{(r)}(\mu) R_\mu.$$

Covariance and the output twirl imply that

$$\mathcal{M}_n(B_n) = \bigoplus_{\substack{\nu \vdash m_n \\ \ell(\nu) \leq d}} z_\nu \frac{I_{\mathcal{H}_\nu^{(d)}}}{D_d^\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad z_\nu \geq 0, \quad \sum_{\nu} z_\nu = \text{Tr } B_n.$$

Fidelity increases in either positive argument: $A \leq B$ implies $\text{Tr} \sqrt{\sqrt{T} A \sqrt{T}} \leq \text{Tr} \sqrt{\sqrt{T} B \sqrt{T}}$. Positivity of $\mathcal{M}_n$ and the direct-sum formula therefore give

$$\begin{aligned} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) &\leq F(\mathcal{M}_n(B_n), (P/r)^{\otimes m_n}) \\ &= \sum_{\ell(\nu) \leq r} \sqrt{\frac{z_\nu \mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu}}. \end{aligned}$$

By Cauchy–Schwarz and the trace identity for B_n ,

$$F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \leq \left[\left(\sum_{\mu} \mathbf{P}_n^{(r)}(\mu) R_\mu \right) \left(\sum_{\nu} \frac{\mathbf{P}_{m_n}^{(r)}(\nu)}{R_\nu} \right) \right]^{1/2}. \quad (68)$$

Both sums run over partitions with at most r rows; output mass in longer diagrams can only strengthen the inequality.

For the asymptotics, put $q = r(d-r)$ and $c_{r,d} = \prod_{i \leq r < j \leq d} [r(j-i)]^{-1}$. Since $\mathbf{P}_N^{(r)}$ is the Young law of the flat spectrum $(1/r, \dots, 1/r)$, [Lemma 2.1](#) in dimension r shows that $\|\lambda/N - \mathbf{1}_r/r\|_\infty \leq N^{-1/3}$ outside an event of probability at most $e^{-N^{1/3}/4}$. On this typical set, (53) gives $R_\lambda = c_{r,d} N^q (1 + O(N^{-1/3}))$ uniformly, while globally $1 \leq R_\lambda \leq (N+1)^q$. Since $N^q e^{-N^{1/3}/4} \rightarrow 0$, it follows that

$$\mathbb{E}_{\mathbf{P}_N^{(r)}} R_\lambda \sim c_{r,d} N^q, \quad \mathbb{E}_{\mathbf{P}_N^{(r)}} \frac{1}{R_\lambda} \sim c_{r,d}^{-1} N^{-q}. \quad (69)$$

The right-hand side of (68) depends on neither the cloner nor P , so substitution yields

$$\limsup_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{P \in \text{Gr}(r,d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \leq \gamma^{-r(d-r)/2}, \quad (70)$$

which completes the converse and the proof of [Theorem 1.3](#).

5 Discussion

5.1 Dependence on spectral mixedness

For fixed $\gamma > 1$, the product formula shows how fidelity changes with spectral mixedness:

$$f'_\gamma(q) = \frac{(\gamma - 1)^2}{2\sqrt{q(\gamma - 1 + q)}(\sqrt{\gamma(\gamma - 1 + q)} + \sqrt{q})^2} > 0, \quad 0 < q < 1.$$

Consequently, if $p, p' \in \Delta_d^\circ$ satisfy

$$\frac{p'_j}{p'_i} \geq \frac{p_j}{p_i} \quad \text{for every } i < j, \quad (71)$$

that is, if every logarithmic spectral gap contracts in passing from p to p' , then

$$F_{\text{orb}}(\gamma, p') \geq F_{\text{orb}}(\gamma, p), \quad F_{\text{univ}}(\gamma, p') \geq F_{\text{univ}}(\gamma, p), \quad (72)$$

with strict inequality if any ratio is strictly larger. Condition (71) implies that p' is majorized by p , but majorization alone does not ensure larger fidelity when $d \geq 3$. For example, $p' = (0.8, 0.16, 0.04)$ is majorized by $p = (0.9, 0.06, 0.04)$, yet $F_{\text{orb}}(10, p') \approx 0.0772 < 0.0791 \approx F_{\text{orb}}(10, p)$: the decrease in the (2, 3) factor outweighs the increases in the other two.

5.2 Fixed-gain comparison with PCT

For a rank bound r , purify-clone-trace (PCT) applies random purification [TWZ25], Werner's pure-state cloner in dimension dr [Wer98], and partial trace over the purifying registers [LTHC26, FGSS26, JSO26]. Denote its output on $\rho^{\otimes n}$ by $\sigma_{n,m}^{\text{PCT}}(\rho)$. For every ρ of rank at most r , the published data-processing guarantee is

$$F(\sigma_{n,m}^{\text{PCT}}(\rho), \rho^{\otimes m}) \geq \left(\frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}} \right)^{1/2} \longrightarrow \gamma^{-(dr-1)/2} \quad (m = m_n). \quad (73)$$

The bound is exact for each purified component before averaging and discarding the purifying registers, but is only a lower bound for the final output. Appendix A analyzes that output directly. The comparisons below use root fidelity and are illustrated in Fig. 3.

Simple full-rank spectra. Fix $p_1 > \dots > p_d > 0$. PCT uses rank bound $r = d$ and purification dimension d^2 ; neither it nor $\mathcal{C}_n^{\text{univ}}$ is supplied p . Write $\sigma_{n,m}^{\text{PCT}}(p)$ for its output on $\rho_p^{\otimes n}$. Theorem A.2 shows that

$$\lim_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) = F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p), \quad (74)$$

where, with $q_{ij} = p_j/p_i$,

$$F_{\text{PCT}}(\gamma, p) = \left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(d-1)/2} \prod_{i < j} \frac{\sqrt{\gamma + (\gamma - 1)q_{ij}} + \sqrt{q_{ij}(\gamma - 1 + \gamma q_{ij})}}{\gamma(1 + q_{ij})}. \quad (75)$$

By Theorem 1.2(a), $F_{\text{univ}}(\gamma, p)$ is the exact limiting fidelity of $\mathcal{C}_n^{\text{univ}}$. Each factor is strictly smaller for PCT: its output multiplies the spectral covariance by $2\gamma - 1$ rather than γ , and each oscillator mode carries more noise than the least-noise amplifier adds.

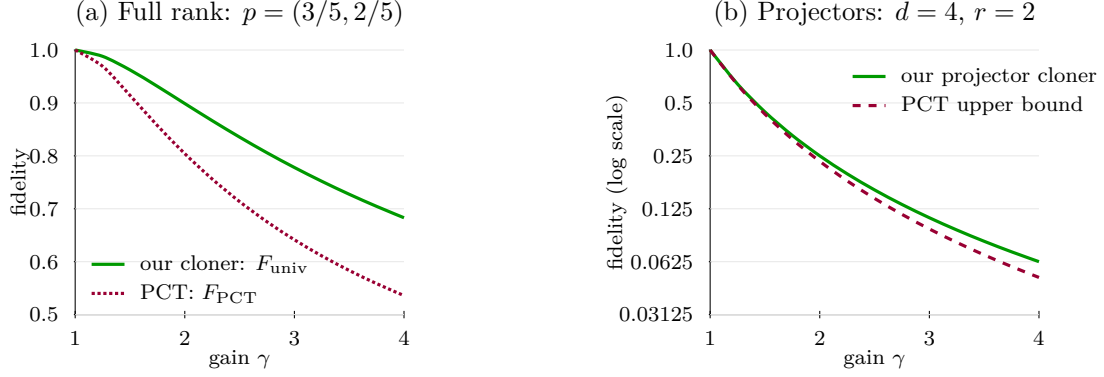


Figure 3: Fixed-gain comparison with PCT. (a) For $p = (3/5, 2/5)$, both curves are exact asymptotic fidelities for protocols without the spectrum (Theorems 1.2 and A.2). (b) For rank-two projectors in $d = 4$, the solid curve is our cloner's exact fidelity γ^{-2} ; the dashed curve is the PCT upper bound $\gamma^{-2}(\sqrt{2\gamma-1}/\gamma)^{1/2}$ (Theorem 1.3 and Proposition A.3). Panel (b) uses a logarithmic fidelity axis.

Projector states. For $\rho = P/r$ with $P \in \text{Gr}(r, d)$ and $1 \leq r < d$, write $\sigma_{n,m}^{\text{PCT}}(P)$ for the output of PCT with rank bound r on $(P/r)^{\otimes n}$. Proposition A.3 gives

$$\limsup_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n}) \leq \gamma^{-r(d-r)/2} \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(r-1)/2} < \gamma^{-r(d-r)/2} \quad (1 < r < d), \quad (76)$$

whereas Theorem 1.3 shows that $\mathcal{C}_n^{\text{proj}}$ attains $\gamma^{-r(d-r)/2}$. The extra factor comes from the computational-basis count statistics: PCT multiplies their limiting covariance by $2\gamma - 1$, even though the flat spectrum is fixed. For $r = 1$, both protocols reduce to Werner's optimal pure-state cloner.

Worst-case cloning over all states. At fixed gain $\gamma > 1$, the optimal fidelity guaranteed by a single channel over all d -dimensional states, without spectral or rank information, is

$$w_n^{\text{all}} = \sup_{\mathcal{M}_n} \inf_{\substack{\rho \geq 0 \\ \text{Tr } \rho = 1}} F(\mathcal{M}_n(\rho^{\otimes n}), \rho^{\otimes m_n}). \quad (77)$$

The PCT guarantee gives $\liminf_n w_n^{\text{all}} \geq \gamma^{-(d^2-1)/2}$. Restricting to the compact simple-spectrum chambers $\Delta_{d,\kappa}$, applying (8), and then letting $\kappa \downarrow 0$ gives

$$\limsup_{n \rightarrow \infty} w_n^{\text{all}} \leq \inf_{p \in \Delta_d^{\circ}} F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) \gamma^{-d(d-1)/4}. \quad (78)$$

For the equality, each orbital factor is at least $f_\gamma(0) = \gamma^{-1/2}$; the geometric spectra $p_i(t) = t^{i-1} / \sum_{j=1}^d t^{j-1}$, $0 < t < 1$, approach this bound as $t \downarrow 0$, since $p_j(t)/p_i(t) = t^{j-i}$. At large gain, these lower and upper bounds scale respectively as $\gamma^{-(d^2-1)/2}$ and $\gamma^{-(d^2-1)/4}$, up to constants, leaving a factor of two between the polynomial exponents. The exact all-state minimax value and whether PCT attains it remain open.

5.3 Small-error sample complexity

To match the sample-complexity literature, this subsection uses squared fidelity, N inputs, and M *additional* outputs, so that

$$m = N + M, \quad \delta = \frac{M}{N}, \quad \gamma = 1 + \delta. \quad (79)$$

For $1 \leq r \leq d$ and sufficiently small $\epsilon > 0$, the concurrent papers [FGSS26, JSO26] prove that cloning with squared fidelity at least $1 - \epsilon$ for every state of rank at most r has optimal worst-case sample complexity

$$N = \Theta\left(\frac{Mrd}{\epsilon}\right). \quad (80)$$

Their upper bounds use PCT, while their lower bounds use flat projectors of rank $\ell = \Theta(r)$, possibly $\ell < r$; in particular, for $r = d$ the hard family is not just I_d/d .

Our limiting fidelities give sharper constants on the specified families, but only in the iterated limit: first $N, M \rightarrow \infty$ at fixed $\delta > 0$, then $\delta \downarrow 0$. We have not controlled approximation errors uniformly as $\gamma_n \downarrow 1$, so the resulting thresholds are not finite-sample bounds in a joint high-fidelity regime and do not improve the worst-case $1/\epsilon$ scaling over the rank-bounded family. The PCT sufficient condition (84) below is the only finite-sample statement here.

Simple full-rank spectra. For fixed p and $\bullet \in \{\text{univ}, \text{PCT}\}$, expand the squared limiting fidelity at $\gamma = 1 + \delta$ and invert it at $1 - \epsilon$. The resulting asymptotic threshold satisfies

$$F_\bullet(1 + \delta, p)^2 = 1 - A_\bullet(p)\delta^2 + O_p(\delta^3), \quad \frac{N}{M} \sim \sqrt{\frac{A_\bullet(p)}{\epsilon}}, \quad (81)$$

where

$$A_{\text{univ}}(p) = \frac{d-1}{8} + \frac{1}{4} \sum_{i < j} \frac{p_i}{p_j}, \quad A_{\text{PCT}}(p) = \frac{d-1}{2} + \frac{1}{4} \sum_{i < j} \frac{(p_i + p_j)^2}{p_i p_j}. \quad (82)$$

Thus $A_{\text{PCT}}(p) > A_{\text{univ}}(p)$, giving our cloner the strictly smaller leading constant. Both coefficients diverge as an eigenvalue approaches zero; the fixed-spectrum $1/\sqrt{\epsilon}$ law is therefore consistent with the worst-case $1/\epsilon$ scaling over the rank-bounded family.

Projector states. For $1 < r < d$, inverting our limiting squared fidelity in the same iterated sense gives

$$\frac{N}{M} = \frac{1}{(1 - \epsilon)^{-1/[r(d-r)]} - 1} \sim \frac{r(d-r)}{\epsilon}. \quad (83)$$

The published guarantee (73) certifies PCT squared fidelity at least $1 - \epsilon$ whenever

$$\frac{N}{M} \geq \frac{1}{(1 - \epsilon)^{-1/(rd-1)} - 1} \sim \frac{rd-1}{\epsilon}, \quad (84)$$

since, with $n = N$ and $m = N + M$, its binomial ratio is $\prod_{j=1}^{rd-1} (N+j)/(N+M+j) \geq [N/(N+M)]^{rd-1}$. The coefficient $rd-1$ describes only this sufficient condition. In the iterated limit, both the lim inf and the lim sup of PCT's actual squared infidelity equal $r(d-r)\delta + O_{r,d}(\delta^2)$ (Proposition A.4), so PCT shares the leading sample ratio $N/M \sim r(d-r)/\epsilon$. Together with (76), this shows that our advantage first appears at second order: the asymptotic squared-fidelity gap is of order δ^2 , with its exact coefficient undetermined. Subleading expansions of the optimal and certified thresholds appear in (125).

5.4 Conjectures for degenerate and rank-deficient spectra

We distinguish known-spectrum orbits from local unknown-spectrum models of fixed rank. Suppose a rank- r state ρ has distinct positive eigenvalues $u_1 > \dots > u_s > 0$, with multiplicities $r_1, \dots, r_s$, where $\sum_a r_a = r$ and $\sum_a r_a u_a = 1$. We conjecture that the known-spectrum optimum on its unitary orbit, defined as in (6), is

$$F_{\text{orb}}(\gamma, \rho) = \gamma^{-r(d-r)/2} \prod_{a < b} f_\gamma\left(\frac{u_b}{u_a}\right)^{r_a r_b}. \quad (85)$$

Rotations within degenerate eigenspaces are stabilizers and do not contribute. Since $f_\gamma(0) = \gamma^{-1/2}$, the first factor accounts for the $r(d-r)$ support–kernel directions. The simple full-rank case $r = d$, $s = d$, $r_a = 1$ recovers (5); the flat projector case $s = 1$ recovers Theorem 1.3, including Werner’s pure-state value at $r = 1$. The exact compression identity in Lemma 4.1 establishes this support–kernel factorization for our sector channel, but does not by itself prove the conjecture for the remaining spectra.

The known-spectrum optimum need not be continuous at a degeneracy, because the orbit loses dimensions. For example, for qubits,

$$\lim_{\substack{p \rightarrow (1/2, 1/2) \\ p_1 > p_2}} F_{\text{orb}}(\gamma, p) = \frac{2\sqrt{\gamma}}{1 + \gamma} < 1,$$

whereas the known-spectrum family at $p = (1/2, 1/2)$ contains only $I_2/2$ and can be cloned perfectly. For each fixed n , the constant channel preparing $(I_2/2)^{\otimes m_n}$ has fidelity tending to one as $p \rightarrow (1/2, 1/2)$, uniformly over the orbit. Thus the degeneracy and large-sample limits do not commute.

For the local unknown-state problem, consider the smooth manifold of rank- r states

$$\mathcal{D}_r^\circ := \{\sigma \succeq 0 : \text{Tr } \sigma = 1, \text{ rank } \sigma = r\}. \quad (86)$$

If P is the support projection of ρ , its tangent space is

$$T_\rho \mathcal{D}_r^\circ = \{X = X^* : \text{Tr } X = 0, (I - P)X(I - P) = 0\}. \quad (87)$$

Let Exp_ρ be the exponential map for the induced Hilbert–Schmidt metric. For fixed L and sufficiently large n , set $\rho_{n,X} = \text{Exp}_\rho(X/\sqrt{n})$ and define

$$v_{n,L}^{(r)}(\rho) := \sup_{\mathcal{M}_n} \inf_{\substack{X \in T_\rho \mathcal{D}_r^\circ \\ \|X\|_2 \leq L}} F\left(\mathcal{M}_n(\rho_{n,X}^{\otimes n}), \rho_{n,X}^{\otimes m_n}\right). \quad (88)$$

The channel may depend on the fixed center ρ , but not on the unknown displacement X . We conjecture that the $\limsup$ and $\liminf$ of $v_{n,L}^{(r)}(\rho)$, first as $n \rightarrow \infty$ and then as $L \rightarrow \infty$, agree. Writing their common value as $F_{\text{univ}}^{(r)}(\gamma, \rho)$, the conjectured formula is

$$F_{\text{univ}}^{(r)}(\gamma, \rho) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma}\right)^{(\sum_{a=1}^s r_a^2 - 1)/2} F_{\text{orb}}(\gamma, \rho). \quad (89)$$

The classical exponent is half the real dimension $\sum_a r_a^2 - 1$ of Hermitian matrices block diagonal in the positive eigenspaces and with zero total trace. The support–kernel directions are already included in $F_{\text{orb}}(\gamma, \rho)$, and the degenerate zero block contributes no additional parameters when the

rank is fixed. Gaussian matrix fluctuations suggest a classical limit for the within-block directions [Kup02, Joh01]. For simple positive spectra (every $r_a = 1$), the fixed-rank LAN theorem of [LN24, Theorem 3.3] has exactly $r - 1$ classical Gaussian parameters, $r(r - 1)/2$ shifted thermal modes, and $r(d - r)$ shifted pure modes, matching the exponent $(r - 1)/2$ here and the factors in (85).

The formula is consistent under positive-eigenvalue collisions at fixed rank: each colliding pair contributes $f_\gamma(1) = 2\sqrt{\gamma}/(1 + \gamma)$, and $(r - 1)/2 + \sum_a \binom{r_a}{2} = (\sum_a r_a^2 - 1)/2$. When also $r = d$ and the spectrum is simple, the formula reduces to the full-rank value $F_{\text{univ}}(\gamma, p)$ in (5).

At a flat projector state, $s = 1$, the local model still allows the positive eigenvalues to vary, unlike the projector family in Theorem 1.3, where they are fixed at $1/r$. This uncertainty gives the conjectured classical factor for $r > 1$; the two values coincide for $r = 1$. Neither model allows rank changes.

Conjectured comparison with PCT. We further conjecture that, at every fixed $\gamma > 1$ and with the same prior spectral or rank information available to both protocols, Schur–Cartan cloners with transition rules adapted to degenerate or rank-deficient spectra outperform PCT on the corresponding families, and that a single spectrum-independent extension strictly beats PCT with rank bound d in the all-state problem (77). The latter would require control near the boundary of the spectral simplex, beyond the pointwise comparisons proved here.

5.5 Further open problems

Beyond the fixed-rank conjectures, the main open questions are uniform control near rank changes, the all-state minimax value (77), finite-copy optimality of our channels, and approximation-error bounds as $\gamma_n \downarrow 1$ on the scale of the vanishing infidelity. The last would turn the iterated small-error comparisons of Section 5.3 into joint high-fidelity bounds.

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Generative AI usage

The original idea for the cloner construction is due to the authors. We initially conjectured its asymptotic optimality but did not have a complete proof. We used OpenAI’s ChatGPT (Sol 5.6 and 6) and Anthropic’s Claude (Fable 5 and 5.1) to assist with the analysis: the evaluation of the cloning fidelity and the converse argument benefited from their suggestions. The same tools assisted with language editing, manuscript organization, and figure preparation. All AI-assisted material was reviewed and revised by the authors, who take full responsibility for the accuracy and content of the manuscript.

A Comparison with purify–clone–trace

We compute the exact asymptotic PCT fidelity at simple full-rank spectra and derive a strict upper bound for flat projector states with $1 < r < d$, proving the comparisons stated in Section 5.

A.1 Common preliminaries

For $D \geq 1$, write $d_D(N) = \dim \text{Sym}^N(\mathbb{C}^D) = \binom{N+D-1}{D-1}$, and let $\Pi_{\text{sym},N}^{(D)}$ project onto this symmetric subspace. On inputs supported on $\text{Sym}^n(\mathbb{C}^D)$, Werner's pure-state cloner [Wer98] is

$$\mathcal{W}_{n,m}^{(D)}(X) = \frac{d_D(n)}{d_D(m)} \Pi_{\text{sym},m}^{(D)}(X \otimes I_D^{\otimes(m-n)}) \Pi_{\text{sym},m}^{(D)}, \quad m \geq n. \quad (90)$$

If ρ has rank at most r , choose a purification $\psi \in \mathbb{C}^d \otimes E$, $E = \mathbb{C}^r$. All purifications in this space differ by a unitary on E . Covariance of Werner's cloner and invariance of partial trace therefore remove the random-purification Haar average:

$$\sigma_{n,m}^{\text{PCT}}(\rho) = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(dr)}(|\psi\rangle\langle\psi|^{\otimes n}). \quad (91)$$

Occupation and coherent-state limits. Fix $D \geq 2$, a unit vector $\psi \in \mathbb{C}^D$, and an orthonormal basis $e_1, \dots, e_{D-1}$ of $K = \psi^\perp$. For $\mathbf{k} = (k_1, \dots, k_{D-1})$, let $|\mathbf{k}\rangle_m$ be the normalized symmetric vector with k_j particles in e_j and the remaining $m - |\mathbf{k}|$ in ψ .

Lemma A.1 (Werner occupation law). *The output $\mathcal{W}_{n,m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n})$ is diagonal in this occupation basis, with eigenvalue*

$$\pi_{n,m}^{(D)}(\mathbf{k}) = \frac{\binom{m-|\mathbf{k}|}{n}}{\binom{m+D-1}{n+D-1}}, \quad |\mathbf{k}| \leq m - n. \quad (92)$$

If $m_n/n \rightarrow \gamma > 1$, its pullback ω_{n,m_n} to the symmetric Fock space $\Gamma_s(K)$ over K , with occupation basis $|\mathbf{k}\rangle$, satisfies

$$\left\| \omega_{n,m_n} - \tau_{(\gamma-1)/\gamma}^{\otimes(D-1)} \right\|_1 \rightarrow 0. \quad (93)$$

Proof. On an occupation vector with $t = |\mathbf{k}|$, the expectation of $\Pi_{\text{sym},m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n} \otimes I) \Pi_{\text{sym},m}^{(D)}$ is $\binom{m-t}{n} / \binom{m}{n}$. Multiplying by $d_D(n)/d_D(m)$ gives (92). The stabilizer $U(D-1)$ acts irreducibly on each $\text{Sym}^t(K)$, and its central phase eliminates cross- t blocks, so these are all the eigenvalues. For each fixed $\mathbf{k}$,

$$\pi_{n,m_n}^{(D)}(\mathbf{k}) \rightarrow \gamma^{-(D-1)} \left(\frac{\gamma-1}{\gamma} \right)^{|\mathbf{k}|}.$$

The limit is a normalized product of geometric laws. The discrete Scheffé lemma therefore gives ℓ^1 convergence, which is (93). $\square$

Let R_M project onto Fock occupations of total number at most M , and let $J_M|\mathbf{k}\rangle = |\mathbf{k}\rangle_M$ on this subspace. The trace-preserving occupation embedding is

$$\mathcal{J}_M(X) = J_M R_M X R_M J_M^* + \text{Tr}[(I - R_M)X] |\psi\rangle\langle\psi|^{\otimes M}. \quad (94)$$

For $Z \in K$, distinguish its coherent Fock vector $|\text{coh}(Z)\rangle$ from the one-particle vector $|Z\rangle$, and put

$$|\psi_{Z,M}\rangle = \frac{|\psi\rangle + M^{-1/2}|Z\rangle}{\sqrt{1 + \|Z\|^2/M}}.$$

For fixed Z ,

$$\left\| \mathcal{J}_M(|\text{coh}(Z)\rangle\langle\text{coh}(Z)|) - |\psi_{Z,M}\rangle\langle\psi_{Z,M}|^{\otimes M} \right\|_1 \rightarrow 0. \quad (95)$$

For $Z \neq 0$, both vectors have excitations only in the direction $Z/\|Z\|$ (the case $Z = 0$ is immediate). With $t = \|Z\|^2$, their number amplitudes after pulling back the product vector by J_M^* are the square roots of

$$B_M(j) = \binom{M}{j} \frac{(t/M)^j}{(1+t/M)^M}, \quad P(j) = e^{-t} \frac{t^j}{j!}.$$

Pointwise convergence and the discrete Scheffé lemma give

$$\sum_{j \geq 0} (\sqrt{B_M(j)} - \sqrt{P(j)})^2 \leq \sum_{j \geq 0} |B_M(j) - P(j)| \rightarrow 0.$$

Thus the pure states converge in trace norm; contractivity of $\mathcal{J}_M$ proves (95).

For $s > 0$, let μ_s be the circular complex Gaussian measure on K with covariance sI_K , so each orthonormal coordinate has $\mathbb{E}|Z_j|^2 = s$. Gaussian integration in the number basis gives the positive coherent-state representation

$$\tau_{s/(1+s)}^{\otimes(D-1)} = \int_K |\text{coh}(Z)\rangle\langle\text{coh}(Z)| d\mu_s(Z), \quad d\mu_s(Z) = \frac{e^{-\|Z\|^2/s}}{(\pi s)^{D-1}} dZ. \quad (96)$$

Specifically, the integral is diagonal with eigenvalue $s^{|k|}/(1+s)^{|k|+D-1}$. Combining (91), (93), and (95) therefore gives, for any fixed purification $\psi \in \mathbb{C}^d \otimes E$,

$$\left\| \sigma_{n,m_n}^{\text{PCT}}(\rho) - \int_K \rho_{Z,m_n}^{\otimes m_n} d\mu_{\gamma-1}(Z) \right\|_1 \rightarrow 0, \quad \rho_{Z,M} = \text{Tr}_E |\psi_{Z,M}\rangle\langle\psi_{Z,M}|. \quad (97)$$

Here the finite Werner output equals $\mathcal{J}_{m_n}(\omega_{n,m_n})$. Contractivity controls its replacement by the thermal state, and dominated convergence controls the coherent-state integral, whose trace-distance integrand is bounded by 2. No uniform estimate over unbounded Z is needed.

A.2 Simple full-rank spectra

Fix $p \in \Delta_d^\circ$, so that $\rho_p = \text{diag}(p_1, \dots, p_d)$ with $p_1 > \dots > p_d > 0$, and recall $q_{ij} = p_j/p_i$. Put

$$h_q(x) = \frac{\sqrt{(1-q)(1-x)}}{1 - \sqrt{qx}}, \quad (98)$$

and define

$$q_{ij}^{\text{PCT}} = \frac{\gamma - 1 + \gamma q_{ij}}{\gamma + (\gamma - 1)q_{ij}}. \quad (99)$$

The asymptotic PCT fidelity (75) can equivalently be written as

$$F_{\text{PCT}}(\gamma, p) = \left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(d-1)/2} \prod_{i < j} h_{q_{ij}}(q_{ij}^{\text{PCT}}). \quad (100)$$

PCT here uses rank bound d , hence purification dimension d^2 .

Theorem A.2 (Exact full-rank fidelities and comparison). *For fixed d, p , and $\gamma > 1$ as above,*

$$\lim_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) = F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p) = \lim_{n \rightarrow \infty} F(\mathcal{C}_n^{\text{univ}}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}), \quad (101)$$

with F_{PCT} as in (100) and F_{univ} as in (5).

Proof. The final equality follows from [Theorem 1.2\(a\)](#). For the PCT limit, apply interior LAN to the mixture (97) and compare the resulting factors.

Step 1: The Gaussian output. Choose

$$S = \text{diag}(\sqrt{p_1}, \dots, \sqrt{p_d}), \quad |\psi_p\rangle = \sum_i \sqrt{p_i} |ii\rangle, \quad K_p = \psi_p^\perp.$$

Identify the tangent vector $|Z\rangle = \sum_{i,j} Z_{ij} |ij\rangle$ with its coefficient matrix. Partial trace has differential

$$D_p(Z) = ZS + SZ^*, \quad \rho_{Z,M} = \rho_p + M^{-1/2} D_p(Z) + O_Z(M^{-1}). \quad (102)$$

Its diagonal and orbital LAN coordinates are

$$h_i(Z) = 2\sqrt{p_i} \Re Z_{ii}, \quad \sum_i h_i(Z) = 0, \quad (103)$$

$$\beta_{ij}(Z) = \frac{D_p(Z)_{ji}}{\sqrt{p_i} - p_j} = \frac{\sqrt{p_i} Z_{ji} + \sqrt{p_j} \overline{Z_{ij}}}{\sqrt{p_i} - p_j}, \quad i < j. \quad (104)$$

The derivative of the local chart from [Section 3.1](#) is $(u, w) \mapsto \text{diag}(u) + [Y_p(w), \rho_p]$, a real linear isomorphism onto the traceless Hermitian matrices. The inverse function theorem and (102) give exact LAN coordinates:

$$\rho_{Z,M} = g_{M,z_M}^{(p)} \text{diag}(p + M^{-1/2} h_M) (g_{M,z_M}^{(p)})^*, \quad (h_M, z_M) = (h(Z), \beta(Z)) + O_Z(M^{-1/2}).$$

Let T_M^p, S_M^p be the forward and reverse interior LAN channels centered at ρ_p , in the normalization of (45), and set

$$\Theta(Z) = \mathbf{N}_{h(Z), \Sigma_p} \otimes \bigotimes_{i < j} D(\beta_{ij}(Z)) \tau_{q_{ij}} D(\beta_{ij}(Z))^*. \quad (105)$$

The bounded-parameter LAN estimates (46)–(47) and trace-norm continuity of Gaussian shifts imply, for each fixed Z ,

$$\left\| T_M^p(\rho_{Z,M}^{\otimes M}) - \Theta(Z) \right\|_1 \rightarrow 0, \quad \left\| \rho_{Z,M}^{\otimes M} - S_M^p(\Theta(Z)) \right\|_1 \rightarrow 0.$$

To average $\Theta(Z)$ under $Z \sim \mu_s$, $s = \gamma - 1$, use the orthogonal decomposition:

$$K_p = \left\{ \sum_i z_i |ii\rangle : \sum_i \sqrt{p_i} z_i = 0 \right\} \oplus \bigoplus_{i < j} \text{span}\{|ij\rangle, |ji\rangle\}$$

The diagonal and pair variables are independent, and (103)–(104) give

$$\text{Cov}(h) = 2s\Sigma_p, \quad \mathbb{E}|\beta_{ij}|^2 = \frac{s(p_i + p_j)}{p_i - p_j}, \quad \mathbb{E}\beta_{ij}^2 = 0. \quad (106)$$

Here the diagonal Gaussian has covariance $s(I - \sqrt{p} \sqrt{p}^\top)$; multiplying its real parts by $2\sqrt{p_i}$ gives the first identity on the trace-zero hyperplane. Each β_{ij} is circular complex Gaussian. Averaging its displacement multiplies the Weyl characteristic function by $\exp(-\mathbb{E}|\beta_{ij}|^2 |\xi|^2)$, leaving a thermal state whose mean photon number increases by $\mathbb{E}|\beta_{ij}|^2$:

$$\bar{n}_{ij}^{\text{PCT}} = \frac{p_j + s(p_i + p_j)}{p_i - p_j} = \frac{(\gamma - 1)p_i + \gamma p_j}{p_i - p_j}, \quad (107)$$

whose geometric parameter is (99). Consequently,

$$\Theta_p^{\text{PCT}} := \int_{K_p} \Theta(Z) d\mu_{\gamma-1}(Z) = \mathbf{N}_{0,(2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}}^{\text{PCT}}. \quad (108)$$

Integrating both LAN approximations by dominated convergence and using (97) gives

$$\left\| T_{m_n}^p(\sigma_{n,m_n}^{\text{PCT}}(p)) - \Theta_p^{\text{PCT}} \right\|_1 \longrightarrow 0, \quad (109)$$

$$\left\| \sigma_{n,m_n}^{\text{PCT}}(p) - S_{m_n}^p(\Theta_p^{\text{PCT}}) \right\|_1 \longrightarrow 0. \quad (110)$$

The target has the same two-sided approximations with

$$\Theta_p = \Theta(0) = \mathbf{N}_{0,\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}}. \quad (111)$$

Step 2: Fidelity and strict comparison. Monotonicity under $T_{m_n}^p$ and $S_{m_n}^p$, together with trace-norm continuity, now gives the two-sided bound

$$F(\Theta_p^{\text{PCT}}, \Theta_p) - o(1) \leq F(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) \leq F(\Theta_p^{\text{PCT}}, \Theta_p) + o(1).$$

The limiting fidelity factorizes, with

$$F(\mathbf{N}_{0,(2\gamma-1)\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(d-1)/2} \quad (112)$$

and $F(\tau_{q_{ij}}^{\text{PCT}}, \tau_{q_{ij}}) = h_{q_{ij}}(q_{ij}^{\text{PCT}})$. This proves the exact value (100).

For one mode with $q \in (0, 1)$, the optimal amplifier has mean $\bar{n}^{\text{opt}} = \gamma q / (1 - q) + \gamma - 1$, hence

$$\bar{n}^{\text{opt}} > \frac{q}{1-q}, \quad \bar{n}^{\text{PCT}} - \bar{n}^{\text{opt}} = \frac{(\gamma-1)q}{1-q} > 0.$$

Thus $q < q^{\text{opt}} < q^{\text{PCT}}$, where $q^{\text{opt}} = (\gamma - 1 + q)/\gamma$ and $f_\gamma(q) = h_q(q^{\text{opt}})$. Since

$$\frac{d}{dx} \log h_q(x) = \frac{\sqrt{q} - \sqrt{x}}{2\sqrt{x}(1-x)(1-\sqrt{qx})} < 0, \quad x > q,$$

every PCT quantum factor is strictly smaller. PCT's classical factor is also smaller: $c \mapsto 2\sqrt{c}/(1+c)$ decreases for $c > 1$, and $2\gamma - 1 > \gamma > 1$. Multiplying the factors proves $F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p)$. $\square$

A.3 Projector states

For projector states, we upper-bound PCT's actual fidelity to compare it with the optimum in Theorem 1.3; its published data-processing lower bound does not suffice for this comparison.

Fix $1 \leq r < d$ and $P \in \text{Gr}(r, d)$, choose an isometry $V_P : \mathbb{C}^r \rightarrow \mathbb{C}^d$ with $V_P V_P^* = P$, and put

$$|\Phi_P\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r V_P |i\rangle \otimes |i\rangle, \quad |\Phi_r\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r |ii\rangle.$$

Let $\sigma_{n,m}^{\text{PCT}}(P)$ denote the system output of PCT on $(P/r)^{\otimes n}$, using a purifying register $\mathbb{C}^r$.

Proposition A.3 (PCT is strictly suboptimal for non-pure projector states). *Define*

$$s_{n,m}^{(r,d)} = \frac{\mathbf{d}_{rd}(n)/\mathbf{d}_{rd}(m)}{\mathbf{d}_{r^2}(n)/\mathbf{d}_{r^2}(m)} = \prod_{j=r^2}^{rd-1} \frac{n+j}{m+j}. \quad (113)$$

There is an exact finite-sample factorization

$$F\left(\sigma_{n,m}^{\text{PCT}}(P), (P/r)^{\otimes m}\right) = \sqrt{s_{n,m}^{(r,d)}} F\left(\zeta_{n,m}^{(r)}, (I_r/r)^{\otimes m}\right), \quad (114)$$

where

$$\zeta_{n,m}^{(r)} = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(r^2)}(|\Phi_r\rangle\langle\Phi_r|^{\otimes n}). \quad (115)$$

If $m_n/n \rightarrow \gamma > 1$, then

$$\limsup_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n}\right) \leq \gamma^{-r(d-r)/2} \left(\frac{\sqrt{2\gamma-1}}{\gamma}\right)^{(r-1)/2}. \quad (116)$$

Consequently, for every $1 < r < d$ and fixed $\gamma > 1$, PCT is strictly below the optimal projector-state value (9). For $r = 1$, the internal factor in (114) is one and PCT reduces to Werner's optimal pure-state cloner.

Proof. By (91), use the fixed purification Φ_P . Werner's channel is the one-row Cartan channel, so (55) applies with $\mu = (n)$, $\nu = (m)$, and $V_P \otimes I_r : \mathbb{C}^{r^2} \rightarrow \mathbb{C}^{rd}$. On the symmetric output, its compression projection is $(P \otimes I_r)^{\otimes m}$ and its dimension factor is $s_{n,m}^{(r,d)}$ by (113). Tracing out E^m gives

$$P^{\otimes m} \sigma_{n,m}^{\text{PCT}}(P) P^{\otimes m} = s_{n,m}^{(r,d)} V_P^{\otimes m} \zeta_{n,m}^{(r)} (V_P^*)^{\otimes m}. \quad (117)$$

Compressing to the target support $P^{\otimes m}$ leaves fidelity unchanged; homogeneity then proves (114).

It remains to bound the internal fidelity. Assume $r > 1$, put

$$\mathbf{H}_r = \left\{ h \in \mathbb{R}^r : \sum_{i=1}^r h_i = 0 \right\}, \quad u_r = \frac{1}{r} \mathbf{1}_r, \quad \Sigma_r = (\text{diag}(u_r) - u_r u_r^\top) \Big|_{\mathbf{H}_r},$$

and measure every system output in the computational basis. If $C_m = (C_{m,1}, \dots, C_{m,r})$ is the resulting count vector, set

$$Z_m = \frac{C_m - m u_r}{\sqrt{m}} \in \mathbf{H}_r.$$

Under the target $(I_r/r)^{\otimes m}$, the multinomial central limit theorem gives

$$Z_m \Longrightarrow \mathbf{N}_{0, \Sigma_r}. \quad (118)$$

Under $\zeta_{n,m_n}^{(r)}$, we will show

$$Z_{m_n} \Longrightarrow \mathbf{N}_{0, (2\gamma-1)\Sigma_r}. \quad (119)$$

Apply (97) to the internal state, with condensate Φ_r , dimension r^2 , and $Z \sim \mu_{\gamma-1}$ on $\Phi_r^\perp$. Identify Z with a traceless complex $r \times r$ matrix. Conditional on Z , measuring the system part of the product of the normalized vector $\Phi_r + m^{-1/2}Z$ gives multinomial probabilities

$$\begin{aligned} p_i^{(m)}(Z) &= \frac{r^{-1} + 2\Re Z_{ii}/\sqrt{rm} + m^{-1} \sum_j |Z_{ij}|^2}{1 + m^{-1} \|Z\|^2} \\ &= \frac{1}{r} + \frac{h_i(Z)}{\sqrt{m}} + O_Z(m^{-1}), \quad h_i(Z) = \frac{2\Re Z_{ii}}{\sqrt{r}}. \end{aligned} \quad (120)$$

The multinomial central limit theorem therefore gives the conditional limit $\mathbf{N}_{h(Z), \Sigma_r}$. Since $\text{Tr } Z = 0$, the mean $h(Z)$ belongs to $\mathbf{H}_r$; under $\mu_{\gamma-1}$ it is centered Gaussian, with

$$\text{Cov}(h(Z))_{ij} = \frac{2(\gamma-1)}{r} \left(\delta_{ij} - \frac{1}{r} \right) = 2(\gamma-1)(\Sigma_r)_{ij}.$$

Dominated convergence of the conditional characteristic functions gives the Gaussian limit with covariance $\Sigma_r + 2(\gamma-1)\Sigma_r = (2\gamma-1)\Sigma_r$. The trace-norm approximation (97) and contractivity of measurement transfer this limit to $\zeta_{n,m_n}^{(r)}$, proving (119).

Let $L_n \Rightarrow L$ and $T_n \Rightarrow T$ be the count laws and their Gaussian limits. For any finite partition $\mathcal{P}$ whose boundaries have zero Gaussian measure, coarse-graining and weak convergence give

$$\limsup_n F(L_n, T_n) \leq \sum_{A \in \mathcal{P}} \sqrt{L(A)T(A)}.$$

Along a refining sequence generating the Borel sets, the right-hand side decreases to $F(L, T)$. This upper semicontinuity, fidelity monotonicity under the count measurement, and (118)–(119) give

$$\limsup_{n \rightarrow \infty} F(\zeta_{n,m_n}^{(r)}, (I_r/r)^{\otimes m_n}) \leq \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(r-1)/2}. \quad (121)$$

Finally, (113) gives

$$s_{n,m_n}^{(r,d)} \longrightarrow \gamma^{-r(d-r)}.$$

Combining this with (114) and (121) proves (116). Its right-hand side is strictly smaller than $\gamma^{-r(d-r)/2}$ when $1 < r < d$, because $2\gamma-1 < \gamma^2$. The $r=1$ assertion follows directly from (115). $\square$

Proposition A.4 (PCT's leading projector small-error coefficient). *Fix $1 < r < d$ and $P \in \text{Gr}(r, d)$, put $a = r(d-r)$, and, for each fixed $0 < \delta < 1/(2r)$, take $m_n/n \rightarrow 1 + \delta$. Define*

$$\mathcal{E}_n(\delta) = 1 - F(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n})^2, \quad C_r(\delta) = (1 - 4\delta^2)^{-(r^2-1)/2} - 1.$$

Then

$$1 - (1 + \delta)^{-a} \leq \liminf_n \mathcal{E}_n(\delta) \leq \limsup_n \mathcal{E}_n(\delta) \leq 1 - (1 + \delta)^{-a} + (1 + \delta)^{-a} C_r(\delta). \quad (122)$$

Thus both envelopes equal $a\delta + O_{r,d}(\delta^2)$ as $\delta \downarrow 0$. In this iterated limit, PCT has the same leading small-error coefficient as the optimal projector cloner, despite its strictly smaller fidelity at each fixed gain.

Proof. It suffices to bound the internal squared infidelity in (114). Specialize the mixture (97) to the condensate Φ_r , and write

$$\tilde{\zeta}_M = \int_{\Phi_r^\perp} \rho_{Z,M}^{\otimes M} d\mu_\delta(Z), \quad \rho_{Z,M} = \frac{I_r}{r} + \frac{A_Z}{\sqrt{M}} + O_Z(M^{-1}), \quad A_Z = \frac{Z + Z^*}{\sqrt{r}}.$$

Here Z is a traceless complex $r \times r$ matrix, and $\|\zeta_{n,m_n}^{(r)} - \tilde{\zeta}_{m_n}\|_1 \rightarrow 0$. For independent $Z, W \sim \mu_\delta$, the scaled purity satisfies

$$r^M \text{Tr } \tilde{\zeta}_M^2 = \mathbb{E} \left[(r \text{Tr}(\rho_{Z,M} \rho_{W,M}))^M \right], \\ (r \text{Tr}(\rho_{Z,M} \rho_{W,M}))^M \longrightarrow e^{r \text{Tr}(A_Z A_W)}.$$

To justify exchanging limit and expectation, contraction under partial trace and the pure-state trace-distance formula give

$$\left\| \rho_{Z,M} - \frac{I_r}{r} \right\|_2 \leq \| |\psi_{Z,M}\rangle\langle\psi_{Z,M}| - |\Phi_r\rangle\langle\Phi_r| \|_1 = 2\sqrt{\frac{\|Z\|^2}{M + \|Z\|^2}}.$$

Since both perturbations have trace zero, Cauchy–Schwarz and $1 + x \leq e^x$ yield

$$0 \leq (r \operatorname{Tr}(\rho_{Z,M} \rho_{W,M}))^M \leq e^{4r\|Z\|\|W\|} \leq e^{2r(\|Z\|^2 + \|W\|^2)}.$$

The last expression is integrable when $\delta < 1/(2r)$. In a Hilbert–Schmidt orthonormal basis of traceless Hermitian matrices, the real parts x_j, y_j of the coordinates of Z, W are independent $\mathbf{N}_{0,\delta/2}$ variables and $r \operatorname{Tr}(A_Z A_W) = 4 \sum_{j=1}^{r^2-1} x_j y_j$. Dominated convergence and the Gaussian cross-moment therefore give

$$\lim_{M \rightarrow \infty} r^M \operatorname{Tr} \tilde{\zeta}_M^2 = \prod_{j=1}^{r^2-1} \mathbb{E} e^{4x_j y_j} = (1 - 4\delta^2)^{-(r^2-1)/2} = 1 + C_r(\delta). \quad (123)$$

For any state σ on $(\mathbb{C}^r)^{\otimes M}$,

$$1 - F\left(\sigma, (I_r/r)^{\otimes M}\right)^2 \leq r^M \operatorname{Tr} \sigma^2 - 1.$$

Indeed, apply $(\sqrt{x} - 1)^2 \leq (x - 1)^2$ to the relative eigenvalues $x = r^M \lambda_j(\sigma)$, average with weights r^{-M} , and use $1 - F^2 \leq 2(1 - F)$. Apply this to $\tilde{\zeta}_M$ and use fidelity continuity to transfer the bound to the PCT output:

$$\limsup_n \left[1 - F\left(\zeta_{n,m_n}^{(r)}, (I_r/r)^{\otimes m_n}\right)^2 \right] \leq C_r(\delta). \quad (124)$$

Finally, the exact support factorization and $s_{n,m_n}^{(r,d)} \rightarrow (1 + \delta)^{-a}$ imply (122). Since $C_r(\delta) = 2(r^2 - 1)\delta^2 + O_r(\delta^4)$, the asserted first-order coefficient follows. For any fixed $0 < \eta < 1$ and sufficiently small $\epsilon > 0$, both envelopes lie below ϵ at $\delta = (1 - \eta)\epsilon/a$ and above it at $\delta = (1 + \eta)\epsilon/a$. Letting $\eta \downarrow 0$ gives the leading sample coefficient a in Section 5.3, without assuming convergence of the fidelity at fixed δ . $\square$

The subleading terms in the two explicitly invertible thresholds of Section 5.3 follow, for any $a > 0$, from

$$\frac{1}{(1 - \epsilon)^{-1/a} - 1} = \frac{a}{\epsilon} - \frac{a+1}{2} + O(\epsilon). \quad (125)$$

Taking $a = r(d - r)$ gives the optimal limiting projector threshold (83); taking $a = rd - 1$ gives the published PCT sufficient threshold (84). Proposition A.4 identifies PCT’s actual leading coefficient, but not its subleading threshold terms or exact internal fidelity.

B Exact-weight PBW calculation of the sector fidelity

We prove Proposition 2.3 on fixed-height weight spaces, where normalized PBW vectors become orthonormal and the Cartan inclusion splits lowering operators with asymptotically binomial weights. Uniform character normalization and a bound on the remaining target mass then give the thermal fidelity $F_{\text{orb}}(\gamma, p)$.

Throughout, $O_L(\cdot)$ may depend on L, d, c, C in $cN \leq \Delta_\alpha^\lambda \leq CN$ for $\alpha \in \mathbf{R}_+$, but is uniform in N and all dominant λ satisfying these inequalities.

B.1 PBW vectors on complete weight spaces

Identify the positive roots of $U(d)$ with

$$R_+ = \{e_i - e_j : 1 \leq i < j \leq d\}.$$

For $\alpha = e_i - e_j$, set

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij}, \quad H_\alpha = [E_\alpha, F_\alpha] = E_{ii} - E_{jj}, \quad \Delta_\alpha^\lambda = \lambda_i - \lambda_j.$$

Fix a total order on R_+ . For $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{R_+}$, write

$$F^\mathbf{r} = \prod_{\alpha \in R_+}^< F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_{\alpha} r_\alpha.$$

For a nonnegative simple-root combination η , let $\text{ht}(\eta)$ be its simple-root height and put

$$P(\eta) = \left\{ \mathbf{r} : \sum_{\alpha \in R_+} r_\alpha \alpha = \eta \right\}.$$

The set $P(\eta)$ is finite, and the PBW theorem implies that $\{F^\mathbf{r} v_\lambda : \mathbf{r} \in P(\eta)\}$ spans $\mathcal{H}_\lambda[\lambda - \eta]$ [FH91]. Define

$$\mathcal{K}_{\lambda,L} = \bigoplus_{\text{ht}(\eta) \leq L} \mathcal{H}_\lambda[\lambda - \eta]. \quad (126)$$

Let $P_{\lambda,L}$ denote the orthogonal projection onto $\mathcal{K}_{\lambda,L}$.

Lemma B.1 (PBW Gram matrices at fixed height). *Let $\lambda = \lambda^{(N)}$ range over dominant weights satisfying*

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in R_+)$$

for fixed $0 < c < C < \infty$. Define

$$\tilde{e}_\mathbf{r}^\lambda = \prod_{\alpha \in R_+}^< \frac{F_\alpha^{r_\alpha}}{\sqrt{r_\alpha!} (\Delta_\alpha^\lambda)^{r_\alpha/2}} v_\lambda. \quad (127)$$

For every fixed L , uniformly over $\text{ht}(\eta) \leq L$ and $\mathbf{r}, \mathbf{s} \in P(\eta)$,

$$\langle \tilde{e}_\mathbf{r}^\lambda, \tilde{e}_\mathbf{s}^\lambda \rangle = \delta_{\mathbf{r},\mathbf{s}} + O_L(N^{-1/2}). \quad (128)$$

Consequently, for all sufficiently large N , the full Gram matrix $G_{\lambda,\eta}$ on every such exact-weight block is invertible. If

$$(e_\mathbf{r}^\lambda)_{\mathbf{r} \in P(\eta)} = (\tilde{e}_\mathbf{r}^\lambda)_{\mathbf{r} \in P(\eta)} G_{\lambda,\eta}^{-1/2}, \quad (129)$$

then these vectors form an orthonormal basis of $\mathcal{H}_\lambda[\lambda - \eta]$, and

$$e_\mathbf{r}^\lambda = \tilde{e}_\mathbf{r}^\lambda + O_L(N^{-1/2}) \quad (130)$$

uniformly for $\text{ht}(\eta) \leq L$ and $\mathbf{r} \in P(\eta)$.

Proof. For $\mathbf{r} \in \mathbf{P}(\eta)$, every positive root has positive simple-root height, so $|\mathbf{r}| \leq \text{ht}(\eta)$. Put

$$\widehat{F}_\alpha = \frac{F_\alpha}{\sqrt{\Delta_\alpha^\lambda}}, \quad \widehat{E}_\alpha = \frac{E_\alpha}{\sqrt{\Delta_\alpha^\lambda}}.$$

Each unnormalized Gram entry $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$ is therefore a word of length at most $2L$. Commuting root operators cannot increase the word length, and each root has height at most $d-1$. Hence every intermediate vector lies in

$$\mathcal{K}_{\lambda, R_L}, \quad R_L := 2L(d-1).$$

The normalized root operators are uniformly bounded on each fixed cutoff. Indeed, put $h = \text{ht}(\alpha)$, set $\mathcal{K}_{\lambda, R} = \{0\}$ for $R < 0$, and write

$$M_R = \|\widehat{F}_\alpha|_{\mathcal{K}_{\lambda, R}}\|^2.$$

Since the cutoffs contain complete weight spaces, $\widehat{E}_\alpha : \mathcal{K}_{\lambda, R} \rightarrow \mathcal{K}_{\lambda, R-h}$ is the adjoint of $\widehat{F}_\alpha : \mathcal{K}_{\lambda, R-h} \rightarrow \mathcal{K}_{\lambda, R}$, and hence its squared norm is M_{R-h} . On $\mathcal{K}_{\lambda, R}$, $[\widehat{E}_\alpha, \widehat{F}_\alpha] = H_\alpha / \Delta_\alpha^\lambda = I + O_R(N^{-1})$. Applying this identity to unit vectors gives

$$M_R \leq M_{R-h} + 1 + O_R(N^{-1}) \leq \lfloor R/h \rfloor + 1 + O_R(N^{-1}) = O_R(1),$$

where the second inequality follows by induction from $M_R = 0$ for $R < 0$. This also bounds $\widehat{E}_\alpha$, without using the Gram estimate. All operator norms below are taken after restriction to $\mathcal{K}_{\lambda, R_L}$, with the codomain norm inherited from $\mathcal{H}_\lambda$.

The matrix-unit commutator formula now gives

$$[\widehat{E}_\alpha, \widehat{F}_\beta] = \delta_{\alpha\beta} I + O_L(N^{-1/2}). \quad (131)$$

For $\alpha = \beta$, the left side is $H_\alpha / \Delta_\alpha^\lambda = I + O_L(N^{-1})$ on the cutoff. For $\alpha \neq \beta$, the unnormalized commutator is zero or, up to sign, a root operator E_γ or F_γ . In the nonzero case, the preceding bound gives $\|[\widehat{E}_\alpha, \widehat{F}_\beta]\| = O_L(\sqrt{\Delta_\gamma^\lambda} / \sqrt{\Delta_\alpha^\lambda \Delta_\beta^\lambda}) = O_L(\sqrt{N}/N) = O_L(N^{-1/2})$.

Commute every $\widehat{E}_\alpha$ to the right in $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$. The scalar commutators in (131) contribute only when $\mathbf{r} = \mathbf{s}$, in which case they give $\prod_\alpha r_\alpha!$. Every other term vanishes or contains an $O_L(N^{-1/2})$ remainder. Only $O_L(1)$ terms occur, and all remaining operator factors have norm $O_L(1)$; consequently,

$$\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} \prod_\alpha r_\alpha! + O_L(N^{-1/2}).$$

Dividing by the factorials in (127) proves (128).

The number and sizes of the exact-weight blocks are bounded in terms of L , so $G_{\lambda, \eta} = I + O_L(N^{-1/2})$ in operator norm. It is therefore invertible for large N , and the PBW spanning vectors form a basis. Functional calculus gives $G_{\lambda, \eta}^{-1/2} = I + O_L(N^{-1/2})$, proving (129) and (130). $\square$

B.2 Character normalization and fixed-height matrices

For $a = (a_1, \dots, a_d)$, write $p^a = \prod_i p_i^{a_i}$.

Lemma B.2 (Uniform character normalization). *Let $K \Subset \Delta_d^\circ$, let $\delta_N \rightarrow 0$, and suppose $p \in K$, $\lambda \in \mathbf{Y}_N$, and $\|\lambda/N - p\|_\infty \leq \delta_N$. Then, uniformly over this family,*

$$\frac{p^\lambda}{s_\lambda(p)} = \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right) + o(1). \quad (132)$$

Proof. Write the numerator in the Weyl character formula as

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma, \quad A_\sigma = \prod_i p_i^{\lambda_{\sigma(i)} + d - \sigma(i)}.$$

For every nonidentity permutation, strict rearrangement and compactness of K give, uniformly for all sufficiently large N ,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} = \sum_i (p_i - p_{\sigma(i)}) \log p_i + O_K(\delta_N + N^{-1}) \geq c_K > 0.$$

Thus the numerator is $A_{\text{id}}(1 + O_K(e^{-c_K N}))$. Cancelling $\prod_i p_i^{d-i}$ against the Weyl denominator gives

$$s_\lambda(p) = p^\lambda \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right)^{-1} (1 + O_K(e^{-c_K N})).$$

This proves (132). $\square$

Index the number basis of $\mathcal{F}_{\text{orb}}$ by $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbf{R}_+}$, writing $\{|\mathbf{r}\rangle\}$, and define the finite-rank projection

$$P_L^{\mathcal{F}} = \sum_{\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L} |\mathbf{r}\rangle \langle \mathbf{r}|. \quad (133)$$

For every fixed L and all sufficiently large N , each highest weight λ satisfying the gap bounds in Lemma B.1 has the orthonormal basis $e_{\mathbf{r}}^\lambda$ of $\mathcal{K}_{\lambda, L}$ constructed there. Define

$$J_{\lambda, L} : \mathcal{K}_{\lambda, L} \longrightarrow P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}, \quad J_{\lambda, L} e_{\mathbf{r}}^\lambda = |\mathbf{r}\rangle. \quad (134)$$

The common index set makes $J_{\lambda, L}$ unitary on its stated domain.

Proof of Proposition 2.3. Fix K and δ_n as in the proposition, write $m = m_n$, and put $\omega = \nu - \mu$. Since $m/n = \gamma_n \rightarrow \gamma > 1$, we have $m/n - 1 \geq (\gamma - 1)/2$ for all sufficiently large n . Uniformly over the allowed parameters,

$$\left\| \frac{\omega}{m - n} - p \right\|_\infty \leq \frac{m/n + 1}{m/n - 1} \delta_n = o(1).$$

Every coordinate and every adjacent gap is uniformly positive on K . Hence $\omega \vdash m - n$ for all sufficiently large n , and $\nu = \mu + \omega \in \mathbf{Y}_m$. For $\alpha = e_i - e_j$, set

$$s_{\alpha, n} := \frac{\Delta_\alpha^\mu}{\Delta_\alpha^\nu} = \frac{n}{m} \frac{\mu_i/n - \mu_j/n}{\nu_i/m - \nu_j/m} = \frac{1}{\gamma} + o(1), \quad 1 - s_{\alpha, n} = \frac{\Delta_\alpha^\omega}{\Delta_\alpha^\nu}. \quad (135)$$

Consequently, there are constants $0 < c_K < C_K < \infty$, depending only on K and γ , such that, for all large n ,

$$c_K n \leq \Delta_\alpha^\mu, \Delta_\alpha^\nu, \Delta_\alpha^\omega \leq C_K n \quad (\alpha \in \mathbf{R}_+). \quad (136)$$

Let $V = V_{\mu, \omega} : \mathcal{H}_\nu \rightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega$, and write $F_\alpha^{(\lambda)} = d\pi_\lambda(F_\alpha)$ for the action of the root operator on $\mathcal{H}_\lambda$. Intertwining and the highest-weight normalization give

$$VF_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})V.$$

Expanding each $(F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})^{r_\alpha}$ binomially gives the exact identity

$$V\tilde{e}_\mathbf{r}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \tilde{e}_\mathbf{a}^\mu \otimes \tilde{e}_{\mathbf{r}-\mathbf{a}}^\omega, \quad (137)$$

where

$$\beta_{\mathbf{r},\mathbf{a}}^{(n)} = \prod_{\alpha \in \mathbf{R}_+} \sqrt{\binom{r_\alpha}{a_\alpha} s_{\alpha,n}^{a_\alpha} (1 - s_{\alpha,n})^{r_\alpha - a_\alpha}}. \quad (138)$$

If $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$, then (137) has $O_L(1)$ terms, all of height at most L , with $0 \leq \beta_{\mathbf{r},\mathbf{a}}^{(n)} \leq 1$. Since $n, m, m-n$ are uniformly comparable, the gap bounds (136) and Lemma B.1 give $e_\mathbf{b}^\lambda = \tilde{e}_\mathbf{b}^\lambda + O_L(n^{-1/2})$ for $\lambda \in \{\mu, \nu, \omega\}$ and all relevant $\mathbf{b}$. Substituting these relations into (137) and using $\|Vx\| = \|x\|$ gives

$$V e_\mathbf{r}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} e_\mathbf{a}^\mu \otimes e_{\mathbf{r}-\mathbf{a}}^\omega + O_L(n^{-1/2}) \quad (139)$$

uniformly in norm for $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq L$.

Put

$$\sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \quad c_\lambda(p) = \frac{p^\lambda}{s_\lambda(p)}, \quad q_\alpha = \frac{p_j}{p_i} \quad (\alpha = e_i - e_j).$$

For $\lambda \in \{\mu, \nu\}$, the weight of $e_\mathbf{b}^\lambda$ is $\lambda - \sum_\alpha b_\alpha \alpha$, and hence

$$\rho_{p,\lambda} e_\mathbf{b}^\lambda = c_\lambda(p) \prod_\alpha q_\alpha^{b_\alpha} e_\mathbf{b}^\lambda.$$

The sector channel gives the local formula

$$\sigma_{\mu,\nu} = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V^*(\rho_{p,\mu} \otimes I_{\mathcal{H}_\omega}) V.$$

All estimates below are uniform over the allowed family. By Lemma B.2,

$$c_\lambda(p) = \prod_\alpha (1 - q_\alpha) + o(1) \quad (\lambda = \mu, \nu),$$

while the Weyl dimension formula, (136), and (135) give

$$\frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} = \prod_{\alpha=e_i-e_j} \frac{\Delta_\alpha^\mu + j - i}{\Delta_\alpha^\nu + j - i} = \prod_{\alpha \in \mathbf{R}_+} (s_{\alpha,n} + O(n^{-1})) = \gamma^{-r_d} + o(1).$$

Substitute (139) into the sector formula. In the resulting double sum, the weight action and orthonormality force $\mathbf{a} = \mathbf{b}$ in the input factor and then $\mathbf{r} = \mathbf{s}$ in the auxiliary factor. The binomial theorem and $\|\rho_{p,\mu}\| \leq 1$ therefore give, for fixed L ,

$$\langle e_\mathbf{r}^\nu, \sigma_{\mu,\nu} e_\mathbf{s}^\nu \rangle = \delta_{\mathbf{r},\mathbf{s}} \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} c_\mu(p) \prod_{\alpha \in \mathbf{R}_+} (1 - s_{\alpha,n} + s_{\alpha,n} q_\alpha)^{r_\alpha} + O_L(n^{-1/2}), \quad (140)$$

$$\langle e_\mathbf{r}^\nu, \rho_{p,\nu} e_\mathbf{s}^\nu \rangle = \delta_{\mathbf{r},\mathbf{s}} c_\nu(p) \prod_{\alpha \in \mathbf{R}_+} q_\alpha^{r_\alpha}. \quad (141)$$

Since $s_{\alpha,n} = \gamma^{-1} + o(1)$ and $\gamma^{-r_d} \prod_\alpha (1 - q_\alpha) = \prod_\alpha (1 - q_\alpha^{(\gamma)})$, with $q_\alpha^{(\gamma)} = 1 - (1 - q_\alpha)/\gamma$ as in (28), the limiting entries in (140) and (141) are those of the thermal products $\tau_p^{(\gamma)}$ and τ_p of (29), respectively. The cutoff space has fixed finite dimension, so this uniform entrywise convergence gives

$$\left\| J_{\nu,L} P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^\mathcal{F} \tau_p^{(\gamma)} P_L^\mathcal{F} \right\|_1 \longrightarrow 0. \quad (142)$$

$$\left\| J_{\nu,L} P_{\nu,L} \rho_{p,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0. \quad (143)$$

To obtain the fidelity assertions, let $(a_\mu)_\mu$ be a probability distribution supported on labels $\mu \in Y_n$ with $\|\mu/n - p\|_\infty \leq \delta_n$. All such pairs (μ, ν) are compatible for large n , by the first part of the proof. Put

$$\bar{\sigma}_{p,\nu} = \sum_\mu a_\mu \sigma_{\mu,\nu}, \quad \sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}).$$

For fixed L , let $\varepsilon_{n,L} \rightarrow 0$ bound both trace-norm errors in (142) and (143), uniformly over the allowed parameters. Since the output frame $J_{\nu,L} P_{\nu,L}$ is independent of μ , convexity gives the same bound for $\bar{\sigma}_{p,\nu}$, uniformly over all mixture weights.

Next we control the target mass above height L . Since K is compact in Δ_d° , there is $b_K < 1$ such that $q_\alpha \leq b_K$ for every $p \in K$ and every positive root α . Since $\text{ht}(\sum_\alpha r_\alpha \alpha) \leq (d-1)r_d \max_\alpha r_\alpha$ and each occupation under τ_p has geometric tail $\Pr(r_\alpha \geq k) = q_\alpha^k$, a union bound gives

$$t_L := \sup_{p \in K} \text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p] \leq r_d b_K^{\lfloor L/((d-1)r_d) \rfloor + 1} \longrightarrow 0 \quad (L \rightarrow \infty). \quad (144)$$

Taking traces in (143) gives

$$\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}] \leq t_L + \varepsilon_{n,L}. \quad (145)$$

Every $\sigma_{\mu,\nu}$, and hence $\bar{\sigma}_{p,\nu}$, commutes with the diagonal torus, by torus invariance of $\rho_{p,\mu}$ and covariance of $\mathcal{C}_{\mu,\nu}$; so does $\rho_{p,\nu}$. Consequently, $P_{\nu,L}$ reduces both states, and the direct-sum identity splits their fidelity into the sum of the cutoff-block and complementary-block fidelities. Schatten Hölder gives $F(A, C) \leq \sqrt{\text{Tr } A \text{ Tr } C}$, so by (145) and $\text{Tr}[(I - P_{\nu,L}) \bar{\sigma}_{p,\nu}] \leq 1$ the complementary fidelity is at most $\sqrt{t_L + \varepsilon_{n,L}}$. The same decomposition for $F(\tau_p^{(\gamma)}, \tau_p)$ with respect to $P_L^{\mathcal{F}}$ bounds its complementary part by $\sqrt{t_L}$. Finally, fidelity is invariant under the isometry $J_{\nu,L}$, so the continuity estimate (34) and the two $\varepsilon_{n,L}$ -bounds above show that the two cutoff fidelities differ by at most $2\sqrt{\varepsilon_{n,L}}$. Altogether,

$$\left| F(\bar{\sigma}_{p,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \leq 2\sqrt{\varepsilon_{n,L}} + \sqrt{t_L + \varepsilon_{n,L}} + \sqrt{t_L}.$$

First let $n \rightarrow \infty$ at fixed L , then let $L \rightarrow \infty$; the convergence is uniform over $p \in K$, over the admissible labels, and over the weights (a_μ) . Finally,

$$F(\tau_p^{(\gamma)}, \tau_p) = \prod_{i < j} F(\tau_{q_{ij}^{(\gamma)}}, \tau_{q_{ij}}) = F_{\text{orb}}(\gamma, p),$$

by multiplicativity of the trace norm under tensor products, (37) and (40). This proves the mixture assertion of Proposition 2.3; taking a point mass gives the individual-sector assertion. $\square$

C Young local limit and randomized rounding

We prove Proposition 2.4 by embedding Young-label laws in lattice cells. Randomized rounding becomes dilation followed by cell averaging; the Gaussian local limit then gives $F_{\text{cl}}(\gamma, d)$.

Define the half-open fundamental cell

$$\mathbb{C} = \left\{ \sum_{i=1}^{d-1} t_i (e_i - e_d) : -\frac{1}{2} \leq t_i < \frac{1}{2} \right\} \subset \mathbb{H}.$$

Volumes are taken with the induced $(d-1)$ -dimensional Lebesgue measure. The rounding rule (23) is equivalent, outside a null set of cell boundaries, to choosing the unique $\tilde{\nu} \in \mathbf{L}_{m_n}$ for which

$$\gamma_n(\mu + U) \in \tilde{\nu} + \mathbf{C}, \quad U = \sum_{i=1}^{d-1} U_i(e_i - e_d).$$

The first $d-1$ coordinates give the nearest-integer rule, and the last is fixed by the coordinate sum. Since U is uniform on $\mathbf{C}$,

$$D_n(\tilde{\nu} \mid \mu) = \frac{|\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})|}{\gamma_n^{d-1} |\mathbf{C}|}. \quad (146)$$

Extend every label law on $\mathbf{Y}_N$ by zero to $\mathbf{L}_N$.

For $\lambda \in \mathbf{L}_N$, define the recentered and rescaled lattice cell

$$\mathbf{C}_{N,p}(\lambda) = \frac{\lambda - Np + \mathbf{C}}{\sqrt{N}}.$$

For any $R : \mathbf{L}_N \rightarrow \mathbb{R}$ satisfying $\sum_{\lambda \in \mathbf{L}_N} |R(\lambda)| < \infty$, define

$$(\mathcal{E}_{N,p}R)(x) = \frac{N^{(d-1)/2}}{|\mathbf{C}|} R(\lambda) \quad (x \in \mathbf{C}_{N,p}(\lambda)).$$

These cells tile $\mathbf{H}$ with boundaries of zero Lebesgue measure. The map $\mathcal{E}_{N,p}$ preserves ℓ^1 -distance, $\|\mathcal{E}_{N,p}R - \mathcal{E}_{N,p}S\|_1 = \|R - S\|_{\ell^1}$, and, for nonnegative R, S , Hellinger affinity, $\int_{\mathbf{H}} \sqrt{(\mathcal{E}_{N,p}R)(\mathcal{E}_{N,p}S)} \, dx = \sum_{\lambda \in \mathbf{L}_N} \sqrt{R(\lambda)S(\lambda)}$. Each cell has volume $|\mathbf{C}|N^{-(d-1)/2}$.

Lemma C.1 (Uniform Young local limit). *For every compact $K \Subset \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{N,p}P_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (147)$$

Proof. Write $f_{N,p} = \mathcal{E}_{N,p}P_{N,p}$, and fix a ball $B_R = \{x \in \mathbf{H} : \|x\| \leq R\}$. If $\mathbf{C}_{N,p}(\lambda)$ meets B_R , then $x_\lambda = (\lambda - Np)/\sqrt{N}$ stays bounded uniformly in N, p . Since the smallest eigenvalue and the adjacent gaps of $p \in K$ are bounded below, every such λ is a Young diagram for large N . The Frobenius dimension formula and Lemma B.2 give, uniformly on these cells,

$$P_{N,p}(\lambda) = \frac{N! \prod_i p_i^{\lambda_i}}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)} \prod_i \prod_{a=1}^{d-i} \frac{Np_i}{\lambda_i + a} (1 + o(1)).$$

Each factor in the two last products is $1 + O_{K,R}(N^{-1/2})$. Stirling's formula and a Taylor expansion of the first factor therefore yield

$$P_{N,p}(\lambda) = \frac{\exp\left[-\frac{1}{2} \sum_i x_{\lambda,i}^2 / p_i\right]}{(2\pi N)^{(d-1)/2} \sqrt{\prod_i p_i}} (1 + o(1)).$$

The normalization agrees with the Gaussian density on $\mathbf{H}$:

$$|\mathbf{C}| = \sqrt{d}, \quad \det_{\mathbf{H}} \Sigma_p = d \prod_i p_i, \quad \langle x, \Sigma_p^{-1} x \rangle = \sum_i \frac{x_i^2}{p_i} \quad (x \in \mathbf{H}).$$

Indeed, the Gram matrix of $(e_i - e_d)_{i < d}$ has determinant d . The ambient covariance $\text{diag}(p) - pp^\top$ has kernel $\text{span}\{(1, \dots, 1)\}$ and principal cofactors $\prod_i p_i$, giving its restricted determinant. Its

inverse on $\mathbf{H}$ sends x to $\text{diag}(p)^{-1}x$ up to a multiple of $(1, \dots, 1)$, giving the quadratic identity. The cell diameter is $O(N^{-1/2})$, so uniform continuity of the Gaussian densities on bounded sets gives

$$\sup_{p \in K} \sup_{x \in B_R} |f_{N,p}(x) - \varphi_{0,\Sigma_p}(x)| \longrightarrow 0.$$

Both densities have integral one. Consequently,

$$\|f_{N,p} - \varphi_{0,\Sigma_p}\|_1 \leq 2 \int_{B_R} |f_{N,p} - \varphi_{0,\Sigma_p}| dx + 2 \int_{B_R^c} \varphi_{0,\Sigma_p}(x) dx.$$

Let $N \rightarrow \infty$, then $R \rightarrow \infty$. The Gaussian tails vanish uniformly on compact K , proving (147). $\square$

Lemma C.2 (Continuous limit of randomized dilation). *Let $\tilde{Q}_{m_n,p} = P_{n,p}D_n$ denote the output law of the transition rule D_n under input law $P_{n,p}$, before the replacement in (24). For every compact $K \subseteq \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma\Sigma_p}\|_1 \longrightarrow 0. \quad (148)$$

$$\sup_{p \in K} |F(\tilde{Q}_{m_n,p}, P_{m_n,p}) - F_{\text{cl}}(\gamma, d)| \longrightarrow 0. \quad (149)$$

Proof. For $a > 0$ and $f \in L^1(\mathbf{H})$, set

$$(\text{dil}_a f)(y) = a^{-(d-1)/2} f(y/\sqrt{a}),$$

and define the target-cell averaging operator by

$$(\text{Av}_{m_n,p} f)(x) := \frac{m_n^{(d-1)/2}}{|\mathbf{C}|} \int_{\mathbf{C}_{m_n,p}(\nu)} f(y) dy, \quad x \in \mathbf{C}_{m_n,p}(\nu).$$

If $X = (\mu - np + U)/\sqrt{n}$, where $\mu \sim P_{n,p}$ and U is uniform on $\mathbf{C}$, then X has density $\mathcal{E}_{n,p}P_{n,p}$. Since $m_n = \gamma_n n$, the event $\sqrt{\gamma_n}X \in \mathbf{C}_{m_n,p}(\nu)$ is exactly the rounding event $\gamma_n(\mu + U) \in \nu + \mathbf{C}$. Thus the dilated density assigns mass $\tilde{Q}_{m_n,p}(\nu)$ to that target cell, and

$$\text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p}P_{n,p}) = \mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p}. \quad (150)$$

Cell averaging is an L^1 -contraction and dilation an L^1 -isometry. Since $\text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} = \varphi_{0,\gamma_n\Sigma_p}$, (150) yields

$$\|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma\Sigma_p}\|_1 \leq \|\mathcal{E}_{n,p}P_{n,p} - \varphi_{0,\Sigma_p}\|_1 + \|\text{Av}_{m_n,p} \varphi_{0,\gamma_n\Sigma_p} - \varphi_{0,\gamma\Sigma_p}\|_1 + \|\varphi_{0,\gamma_n\Sigma_p} - \varphi_{0,\gamma\Sigma_p}\|_1.$$

The first term tends uniformly to zero by Lemma C.1, and the last does so because $\gamma_n \rightarrow \gamma$. For the middle term, the cellwise L^1 Poincaré inequality gives, for smooth f with integrable gradient,

$$\|\text{Av}_{m_n,p} f - f\|_1 \leq C_d m_n^{-1/2} \|\nabla f\|_1.$$

Gaussian gradients have uniformly bounded L^1 -norm on the compact covariance family $\gamma_n\Sigma_p$, so this term is $O_K(m_n^{-1/2})$. This proves (148).

Applying Lemma C.1 at sample size m_n gives

$$\mathcal{E}_{m_n,p} P_{m_n,p} \longrightarrow \varphi_{0,\Sigma_p}$$

uniformly in L^1 . The embedding preserves Hellinger affinity, and (34) also holds for nonnegative L^1 densities. Therefore

$$F(\tilde{Q}_{m_n,p}, P_{m_n,p}) \longrightarrow F(\mathbf{N}_{0,\gamma\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = F_{\text{cl}}(\gamma, d)$$

uniformly on K . $\square$

Proof of Proposition 2.4. Write $e = \tilde{\nu} - \gamma_n \mu$. The nearest-integer rule gives $|e_i| \leq (\gamma_n + 1)/2$ for $i < d$, and $e_d = -\sum_{i < d} e_i$. Thus $\|e\|_\infty \leq c_{\text{rnd}}$, where

$$c_{\text{rnd}} = \frac{d-1}{2} \left(1 + \sup_n \gamma_n \right) < \infty.$$

For $p \in K$ and $\mu \in \mathbb{T}_n(p)$, since $\tilde{\nu}/m_n = \mu/n + e/m_n$, this gives

$$\left\| \frac{\tilde{\nu}}{m_n} - p \right\|_\infty \leq n^{-1/3} + \frac{c_{\text{rnd}}}{m_n}.$$

The compatibility assertion of Proposition 2.3, with $\delta_n = n^{-1/3} + c_{\text{rnd}}/m_n$, therefore gives $\tilde{\nu} - \mu \vdash m_n - n$ for all large n . No rounded label is changed for typical μ , proving (30).

Coupling the raw and final labels by the fallback rule therefore gives

$$\|Q_{m_n, p} - \tilde{Q}_{m_n, p}\|_{\ell^1} \leq 2P_{n, p}(\mathbb{T}_n(p)^c) = o(1)$$

uniformly on K , by Lemma 2.1. Combining (149) with (34) proves (31). $\square$

D Unified Gaussian flat-prior converse

We prove both converses in Theorem 3.1 by combining weighted fidelity, compact-observable limits, a least-noise moment, and Gaussian translation in Proposition D.4.

The classical parameter space has dimension $k = 0$ in the orbital model and $k = d - 1$ in the full model. All trace-class-valued integrals below are Bochner integrals in trace norm. For classical-quantum maps, complete positivity means positivity of $\text{id}_{M_a} \otimes \Lambda$ for every $a \geq 1$, with the matrix factor acting on the quantum trace-class fibers; the classical algebra is abelian.

D.1 Weighted fidelity and compact limits

Lemma D.1 (Weighted fidelity bound). *Let R, T be positive trace-class operators, and let W be bounded, positive, and injective. Define*

$$\text{Tr}(TW^{-1}) := \lim_{\varepsilon \downarrow 0} \text{Tr}(T(W + \varepsilon I)^{-1}) \in [0, \infty].$$

If this quantity is finite, then

$$F(R, T)^2 \leq \text{Tr}(RW) \text{Tr}(TW^{-1}). \quad (151)$$

Let $\mathbf{E}$ be a finite-dimensional Euclidean space, let $R \in L^1(\mathbf{E}; \mathcal{T}_1(\mathcal{F}_{\text{orb}}))$ be positive, let g be a probability density on $\mathbf{E}$, and let $\chi > 0$ be measurable. For positive R, S in this L^1 -space, set $F(R, S) = \int_{\mathbf{E}} F(R(y), S(y)) \, dy$, extending the hybrid-state convention in Section 3. The notation gT means the density $y \mapsto g(y)T$. Whenever the right-hand side below is finite,

$$F(R, gT)^2 \leq \left(\int_{\mathbf{E}} \chi(y) \text{Tr}[R(y)W] \, dy \right) \left(\int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right) \text{Tr}(TW^{-1}). \quad (152)$$

Proof. For $W_\varepsilon = W + \varepsilon I$, the Schatten Hölder inequality gives

$$F(R, T) = \|(R^{1/2}W_\varepsilon^{1/2})(W_\varepsilon^{-1/2}T^{1/2})\|_1 \leq \sqrt{\text{Tr}(RW_\varepsilon)} \sqrt{\text{Tr}(TW_\varepsilon^{-1})}.$$

Squaring and letting $\varepsilon \downarrow 0$, using $\text{Tr}(RW_\varepsilon) = \text{Tr}(RW) + \varepsilon \text{Tr} R$ and monotone convergence for the second trace, proves (151). For the hybrid state,

$$F(R, gT) = \int_{\mathbb{E}} \sqrt{g(y)} F(R(y), T) dy.$$

Apply the first inequality pointwise and then Cauchy–Schwarz to the factors $\sqrt{\chi(y)} \text{Tr}[R(y)W]$ and $\sqrt{g(y)/\chi(y)}$. This gives (152). $\square$

Lemma D.2 (Compact-observable limits of completely positive maps). *Let $\mathcal{H}_0, \mathcal{H}_1$ be separable Hilbert spaces and let $\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ be completely positive maps with $\sup_j \|\Gamma_j\|_{1 \rightarrow 1} < \infty$. There are a subsequence and a completely positive map $\Gamma : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ such that*

$$\text{Tr}[\Gamma_j(X)K] \longrightarrow \text{Tr}[\Gamma(X)K] \quad (153)$$

for every $X \in \mathcal{T}_1(\mathcal{H}_0)$ and every compact $K \in \mathcal{K}(\mathcal{H}_1)$. If $\limsup_j \text{Tr} \Gamma_j(X) \leq c \text{Tr} X$ for every $X \geq 0$, then $\text{Tr} \Gamma(X) \leq c \text{Tr} X$.

Suppose in addition that $U_g \in \mathcal{U}(\mathcal{H}_0)$ and $V_g \in \mathcal{U}(\mathcal{H}_1)$ are families of unitaries indexed by a set G , and that, for every fixed $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$,

$$\|\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*\|_1 \longrightarrow 0.$$

Then

$$\Gamma(U_g X U_g^*) = V_g \Gamma(X) V_g^* \quad (154)$$

for every $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$.

Proof. Let $M = \sup_j \|\Gamma_j\|_{1 \rightarrow 1}$, and let $\mathcal{D} \subset \mathcal{K}(\mathcal{H}_1)$ be a countable norm-dense set. The predual $\mathcal{T}_1(\mathcal{H}_0)$ is separable, so a diagonal weak-* compactness argument gives a subsequence for which $\Gamma_j^*(K)$ converges weak-* in $\mathcal{B}(\mathcal{H}_0)$ for every $K \in \mathcal{D}$. The bound $\|\Gamma_j^*(K)\| \leq M \|K\|$ extends both the limit and the convergence to every compact K , producing a bounded complex-linear map $\Gamma^* : \mathcal{K}(\mathcal{H}_1) \rightarrow \mathcal{B}(\mathcal{H}_0)$.

For every positive block matrix $[K_{ab}]$ of compact operators, $[\Gamma_j^*(K_{ab})] \geq 0$, so its entrywise weak-* limit $[\Gamma^*(K_{ab})]$ is positive. Thus Γ^* is completely positive.

Since $\mathcal{K}(\mathcal{H}_1)^* = \mathcal{T}_1(\mathcal{H}_1)$, define $\Gamma(X)$ by

$$\text{Tr}[\Gamma(X)K] = \text{Tr}[X\Gamma^*(K)] \quad (K \in \mathcal{K}(\mathcal{H}_1)).$$

Then Γ is completely positive and (153) holds. If $P_N \uparrow I$ are finite-rank projections and $X \geq 0$, then

$$\text{Tr} \Gamma(X) = \sup_N \lim_j \text{Tr}[\Gamma_j(X)P_N] \leq \limsup_j \text{Tr} \Gamma_j(X) \leq c \text{Tr} X.$$

Finally, for fixed g, X and compact K , the operator $V_g^* K V_g$ is compact. Passing to the limit gives

$$\text{Tr}[\Gamma(U_g X U_g^*)K] = \text{Tr}[V_g \Gamma(X) V_g^* K].$$

Compact operators separate trace-class operators, so the limiting map is covariant. $\square$

D.2 The least-noise oscillator moment

Fix $\gamma > 1$. For $s \geq 1$ and $q_1, \dots, q_s \in (0, 1)$, put

$$q_\ell^{(\gamma)} = 1 - \frac{1 - q_\ell}{\gamma}, \quad \tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell^{(\gamma)}}.$$

Fact D.I (Covariant-amplifier representation). *Let $G_1, \dots, G_s > 1$, and write $\sqrt{\mathbf{G}}z = (\sqrt{G_\ell}z_\ell)_{\ell=1}^s$. Every s -mode channel Λ satisfying*

$$\Lambda(D(z)XD(z)^*) = D(\sqrt{\mathbf{G}}z)\Lambda(X)D(\sqrt{\mathbf{G}}z)^*$$

admits an idler realization

$$a_{\ell,\text{out}} = \sqrt{G_\ell} a_{\ell,\text{in}} + \sqrt{G_\ell - 1} b_\ell^*, \quad \ell = 1, \dots, s. \quad (155)$$

The dilation unitary is a product of two-mode squeezers. The joint idler state is uniquely determined by the channel and need not be Gaussian, number diagonal, or a product state.

Proof. Covariance and Weyl irreducibility give $\Lambda^*(D(w)) = f(w)D(\sqrt{\mathbf{G}}w)$. Vacuum matrix elements show that f is continuous at zero, so Λ is a linear bosonic channel. Its defect form is $\bigoplus_{\ell=1}^s (1 - G_\ell)\Omega_2$, where $\Omega_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. This form is invertible, so [LSW18, Lemma 7] gives an exact dilation with s idler modes. Choose the noise matrix $\bigoplus_{\ell=1}^s \sqrt{G_\ell - 1} \text{diag}(1, -1)$ and complete these blocks with a product of two-mode squeezers, yielding (155). The noise matrix is invertible, so f determines the idler characteristic function and hence its state. $\square$

Proposition D.3 (Least-noise moment). *Let Γ be a completely positive trace-nonincreasing map on the s -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

Let $W(\hat{\mathbf{N}}) := \prod_{\ell=1}^s w_\ell(\hat{N}_\ell)$, where every w_ℓ is bounded, nonnegative, and nonincreasing on $\mathbb{Z}_{\geq 0}$. Then

$$\text{Tr}[\Gamma(\tau)W(\hat{\mathbf{N}})] \leq \text{Tr}[\Gamma(\tau)] \text{Tr}[\tau^{(\gamma)}W(\hat{\mathbf{N}})]. \quad (156)$$

Proof. Displacement covariance makes $\Gamma^*(I)$ commute with every Weyl displacement. Irreducibility therefore gives $\Gamma^*(I) = cI$, with $c = \text{Tr} \Gamma(\tau) \in [0, 1]$. The case $c = 0$ is immediate; otherwise $\Lambda = c^{-1}\Gamma$ is a channel with the same covariance.

The quantum-limited amplifier $\mathcal{A}_G$ sends vacuum to $\tau_{1-1/G}$. Thus

$$\tau = \left(\bigotimes_{\ell=1}^s \mathcal{A}_{1/(1-q_\ell)} \right) (|0\rangle\langle 0|). \quad (157)$$

The composite $\mathcal{B} = \Lambda \circ \bigotimes_{\ell=1}^s \mathcal{A}_{1/(1-q_\ell)}$ therefore sends joint vacuum to $\Lambda(\tau)$, with modewise gains

$$G_\ell = \frac{\gamma}{1 - q_\ell}, \quad \lambda_\ell = 1 - G_\ell^{-1} = q_\ell^{(\gamma)}.$$

Apply Fact D.I to $\mathcal{B}$, and let σ be its joint idler state.

For a single two-mode squeezer, normal-ordered disentangling [BR97] gives, with $\lambda = 1 - G^{-1}$,

$$U_G|0, l\rangle = (1 - \lambda)^{(l+1)/2} \sum_{k \geq 0} \sqrt{\binom{k+l}{l}} \lambda^{k/2} |k, k+l\rangle. \quad (158)$$

In the product expansion, the $\mathbf{k}$ -diagonal signal element survives the idler trace only when $\mathbf{k} + \mathbf{l} = \mathbf{k} + \mathbf{l}'$, hence $\mathbf{l} = \mathbf{l}'$. Only the joint diagonal $a_1 = \langle \mathbf{l} | \sigma | \mathbf{l} \rangle$ contributes to the output number probabilities:

$$\langle \mathbf{k} | \Lambda(\tau) | \mathbf{k} \rangle = \sum_{\mathbf{l} \in \mathbb{Z}_{\geq 0}^s} a_1 \prod_{\ell=1}^s \binom{k_\ell + l_\ell}{l_\ell} (1 - \lambda_\ell)^{l_\ell+1} \lambda_\ell^{k_\ell}. \quad (159)$$

Conditional on $\mathbf{l}$, the coordinates are independent $\text{NB}(l_\ell + 1, \lambda_\ell)$ variables; the mixing law a_1 may be correlated. No product or phase-invariance assumption on the idler is needed.

Sample $\mathbf{L}$ with law a_1 and, independently, mutually independent geometric variables $Z_{\ell,j}$ with $\Pr(Z_{\ell,j} = k) = (1 - \lambda_\ell) \lambda_\ell^k$ for $j, k \geq 0$. The coupling

$$N_\ell = \sum_{j=0}^{L_\ell} Z_{\ell,j} \geq Z_{\ell,0} \quad (\ell = 1, \dots, s)$$

has the output law (159), while $(Z_{\ell,0})_\ell$ has the number law of $\tau^{(\gamma)}$. Since each w_ℓ is nonnegative and nonincreasing,

$$\text{Tr}[\Lambda(\tau)W] = \mathbb{E} \prod_{\ell} w_\ell(N_\ell) \leq \mathbb{E} \prod_{\ell} w_\ell(Z_{\ell,0}) = \text{Tr}[\tau^{(\gamma)}W].$$

Multiplying by $c = \text{Tr} \Gamma(\tau)$ proves (156). $\square$

Specialize to $s = r_d$, $q_\ell = q_{ij}$, $\tau = \tau_p$, and $\tau^{(\gamma)} = \tau_p^{(\gamma)}$. For $\mathbf{k} = (k_{ij})_{i < j} \in \mathbb{Z}_{\geq 0}^{r_d}$, let $t_{\mathbf{k}}$ and $t_{\mathbf{k}}^{(\gamma)}$ be the corresponding number-basis eigenvalues, and define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (160)$$

Equivalently,

$$W_\gamma = \prod_{i < j} w_{ij}(\hat{N}_{ij}), \quad w_{ij}(n) = \sqrt{\frac{1 - q_{ij}}{1 - q_{ij}^{(\gamma)}}} \left(\frac{q_{ij}}{q_{ij}^{(\gamma)}} \right)^{n/2}.$$

Here $\hat{N}_{ij}$ is the number operator of the mode indexed by $i < j$, and $|\mathbf{k}| := \sum_{i < j} k_{ij}$ is the total occupation number. Since $q_{ij} < q_{ij}^{(\gamma)}$, each $w_{ij}(n)$ is positive, bounded, and strictly decreasing, and the eigenvalues of W_γ tend to zero as $|\mathbf{k}| \rightarrow \infty$. Thus W_γ is bounded, injective, and compact, with

$$\text{Tr}(\tau_p^{(\gamma)} W_\gamma) = \text{Tr}(\tau_p W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = F_{\text{orb}}(\gamma, p), \quad (161)$$

where the inverse is understood as in Lemma D.1.

D.3 One flat-prior converse for both Gaussian models

Proposition D.4 (Flat-prior converse inequality). *Let $k \in \{0, d-1\}$. For $k = 0$, let $\mathbf{E} = \{0\}$, omit the classical component, and interpret h -integration as evaluation at 0. For $k = d-1$, let $\mathbf{E} = \mathbf{H}$ with its induced Euclidean measure. Let*

$$\mathbf{E}_L = \{h \in \mathbf{E} : \|h\|_\infty \leq L\}, \quad \Xi_L = \mathbf{E}_L \times \mathbf{Z}_L,$$

with $|\mathbf{E}_L| = 1$ when $k = 0$. Define

$$R_{0,z}^{(0)} = \Phi_z^{(p)}, \quad S_{0,z}^{(0)} = \Psi_z^{(p,\gamma)}, \quad R_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{in}}, \quad S_{h,z}^{(d-1)} = \Theta_{h,z}^{\text{tar}}, \quad c_k = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{k/2}.$$

For $k = 0$, the supremum below is over quantum channels on $\mathcal{F}_{\text{orb}}$; for $k = d - 1$, it is over classical-quantum channels. Then

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_{h,z}^{(k)}), S_{h,z}^{(k)}) \, dh \, dz \leq c_k F_{\text{orb}}(\gamma, p). \quad (162)$$

Proof. For $k = 0$, set $g_1 = g_\gamma = \chi = 1$ on the one-point space. For $k = d - 1$, let g_1 and g_γ be the centered Gaussian densities with covariances Σ_p and $\gamma \Sigma_p$, and set

$$\chi(y) = \sqrt{\frac{g_1(y)}{g_\gamma(y)}} = \gamma^{k/4} \exp \left[-\frac{\gamma - 1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle \right].$$

This function is bounded and strictly positive. In both cases,

$$\int_{\mathbf{E}} \frac{g_1(y)}{\chi(y)} \, dy = \int_{\mathbf{E}} \sqrt{g_1(y) g_\gamma(y)} \, dy = c_k. \quad (163)$$

For $k = d - 1$, completing the square also gives

$$H(r) := \int_{\mathbf{E}} g_1(h) \chi(r + \sqrt{\gamma} h) \, dh = c_k \exp \left[-\frac{\gamma - 1}{2\gamma(\gamma + 1)} \langle r, \Sigma_p^{-1} r \rangle \right] \leq c_k. \quad (164)$$

For $\xi = (h, z)$, define

$$(\mathcal{U}_\xi^{\text{in}} R)(x) = D(z) R(x - h) D(z)^*, \quad (\mathcal{U}_\xi^{\text{out}} R)(y) = D(\sqrt{\gamma} z) R(y - \sqrt{\gamma} h) D(\sqrt{\gamma} z)^*.$$

Then $R_\xi^{(k)} = \mathcal{U}_\xi^{\text{in}}(g_1 \tau_p)$ and $S_\xi^{(k)} = \mathcal{U}_\xi^{\text{out}}(g_1 \tau_p)$. For a channel Λ , define its average pointwise by

$$\bar{\Lambda}_L(R) = \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\eta}^{\text{out}}(\Lambda(\mathcal{U}_\eta^{\text{in}} R)) \, d\eta.$$

For each fixed input R , the integrand is trace-norm continuous and bounded in norm by $\|R\|_1$, so the Bochner integral is well-defined. Invariance of F under the isometries $\mathcal{U}_\xi^{\text{out}}$ and joint concavity, applied to the Bochner average, give

$$F_L(\Lambda) = \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_\xi^{(k)}), S_\xi^{(k)}) \, d\xi \leq F(\bar{\Lambda}_L(g_1 \tau_p), g_1 \tau_p). \quad (165)$$

Let $F_* = \limsup_{L \rightarrow \infty} \sup_{\Lambda} F_L(\Lambda)$. Choose $L_j \rightarrow \infty$ with $\sup_{\Lambda} F_{L_j}(\Lambda) \rightarrow F_*$, and channels $\tilde{\Lambda}_j$ satisfying $F_{L_j}(\tilde{\Lambda}_j) \geq \sup_{\Lambda} F_{L_j}(\Lambda) - j^{-1}$. Set

$$\Lambda_j = \overline{\tilde{\Lambda}_{j, L_j}}, \quad R_j = \Lambda_j(g_1 \tau_p).$$

Then

$$F_{L_j}(\tilde{\Lambda}_j) \rightarrow F_*, \quad F_{L_j}(\tilde{\Lambda}_j) \leq F(R_j, g_1 \tau_p). \quad (166)$$

Write

$$A_{j,u} = \mathcal{U}_{-u}^{\text{out}} \circ \tilde{\Lambda}_j \circ \mathcal{U}_u^{\text{in}}.$$

The group law and the change of variables $u = \eta + \xi$ give

$$\begin{aligned}\Lambda_j(\mathcal{U}_\xi^{\text{in}} R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j} + \xi} A_{j,u}(R) \, du, \\ \mathcal{U}_\xi^{\text{out}} \Lambda_j(R) &= \mathcal{U}_\xi^{\text{out}} \frac{1}{|\Xi_{L_j}|} \int_{\Xi_{L_j}} A_{j,u}(R) \, du.\end{aligned}$$

Cancel the intersection of the two averaging sets. Since $\mathcal{U}_\xi^{\text{out}}$ is a trace-norm isometry and each $A_{j,u}$ is a channel, hence a trace-norm contraction,

$$\|\Lambda_j(\mathcal{U}_\xi^{\text{in}} R) - \mathcal{U}_\xi^{\text{out}} \Lambda_j(R)\|_1 \leq b_j(\xi) \|R\|_1, \quad b_j(\xi) := \frac{|(\Xi_{L_j} + \xi) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|}. \quad (167)$$

Here $0 \leq b_j \leq 2$, and $b_j(\xi) = O_\xi(L_j^{-1}) \rightarrow 0$ for fixed ξ , since Ξ_1 is a bounded polytope and $\Xi_L = L\Xi_1$.

Define

$$\Gamma_j^\chi(X) = \int_{\mathbb{E}} \chi(y) [\Lambda_j(g_1 X)](y) \, dy.$$

These maps are completely positive and satisfy $\|\Gamma_j^\chi\|_{1 \rightarrow 1} \leq \|\chi\|_\infty$. We first show the sharper trace bound

$$\text{Tr} \Gamma_j^\chi(X) \leq (c_k + e_j) \text{Tr} X \quad (X \geq 0), \quad e_j := \|\chi\|_\infty \int_{\mathbb{E}} g_1(h) b_j(h, 0) \, dh \rightarrow 0. \quad (168)$$

For $k = 0$, this is trace preservation, with $c_0 = 1$ and $e_j = 0$. For $k = d - 1$, take a nonnegative probability approximate identity u_ε on $\mathbb{E}$ and put $B_{j,\varepsilon} = \Lambda_j(u_\varepsilon X)$, whose total trace is $\text{Tr} X$. Expressing $g_1 * u_\varepsilon$ as an average of translates of u_ε , (167) gives

$$\left\| \Lambda_j((g_1 * u_\varepsilon)X) - \int_{\mathbb{E}} g_1(h) \mathcal{U}_{(h,0)}^{\text{out}} B_{j,\varepsilon} \, dh \right\|_1 \leq \text{Tr} X \int_{\mathbb{E}} g_1(h) b_j(h, 0) \, dh.$$

Pairing with χ , changing variables, and using (164) therefore yields

$$\int_{\mathbb{E}} \chi(y) \text{Tr} [\Lambda_j((g_1 * u_\varepsilon)X)(y)] \, dy \leq \int_{\mathbb{E}} H(r) \text{Tr} B_{j,\varepsilon}(r) \, dr + e_j \text{Tr} X \leq (c_k + e_j) \text{Tr} X.$$

Letting $\varepsilon \downarrow 0$ proves (168), since $g_1 * u_\varepsilon \rightarrow g_1$ in L^1 . Dominated convergence gives $e_j \rightarrow 0$.

Taking $\xi = (0, z)$ in (167) and integrating against χ gives

$$\|\Gamma_j^\chi(D(z)X D(z)^*) - D(\sqrt{\gamma}z) \Gamma_j^\chi(X) D(\sqrt{\gamma}z)^*\|_1 \leq \|\chi\|_\infty b_j(0, z) \|X\|_1 \rightarrow 0.$$

Hence, after passing to a subsequence, Lemma D.2 and (168) give a completely positive, displacement-covariant limit Γ^χ with $\text{Tr} \Gamma^\chi(X) \leq c_k \text{Tr} X$ for $X \geq 0$, such that

$$\text{Tr} [\Gamma_j^\chi(X) K] \rightarrow \text{Tr} [\Gamma^\chi(X) K] \quad (169)$$

for trace-class X and compact K . Since $c_k \leq 1$, the limit is trace nonincreasing. With

$$M_j = \int_{\mathbb{E}} \chi(y) \text{Tr} [R_j(y) W_\gamma] \, dy = \text{Tr} [\Gamma_j^\chi(\tau_p) W_\gamma],$$

compactness of W_γ gives

$$M_j \rightarrow \text{Tr} [\Gamma^\chi(\tau_p) W_\gamma]. \quad (170)$$

Moreover, [Lemma D.1](#), [\(163\)](#), and [\(161\)](#) give

$$F(R_j, g_1 \tau_p)^2 \leq M_j c_k F_{\text{orb}}(\gamma, p). \quad (171)$$

By [Proposition D.3](#),

$$\lim_j M_j = \text{Tr}[\Gamma^\chi(\tau_p) W_\gamma] \leq \text{Tr} \Gamma^\chi(\tau_p) F_{\text{orb}}(\gamma, p) \leq c_k F_{\text{orb}}(\gamma, p).$$

Combining [\(166\)](#) with [\(171\)](#) yields $F_* \leq c_k F_{\text{orb}}(\gamma, p)$, proving both cases. $\square$

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